POLICY ANALYSIS

Building Canada's Early Warning and Response System for AI Job Displacement

A proposal for how Canada can detect AI-driven labour displacement before it reaches crisis scale, and how Mila can shape the national response.

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This report reflects the views of the author and should not be viewed as representing the views of Mila, nor those of Harvard University or any of its faculty.

Executive Summary

Evidence is emerging that generative AI is producing localized labour market disruptions among specific groups of younger workers, even as aggregate Canadian employment remains stable. A January 2026 Statistics Canada report found that employment for 15- to 29-year-olds in highly exposed, coding-intensive professions remained stagnant between late 2022 and late 2025, while employment for workers aged 30 to 49 in the same occupations grew. A November 2025 study by researchers at Stanford's Digital Economy Lab, using ADP payroll microdata combined with Anthropic's AI usage statistics, found a 16% relative employment decline among U.S. workers aged 22 to 25 in occupations where AI was predominantly being used to automate tasks. These disruptions fall below the detection threshold of Canada's standard government surveys. The Labour Force Survey lacks the cell sizes to slice by occupation, age, and province simultaneously, and the Census runs on a five-year cycle. Canada's labour market institutions do not have the capacity to detect an entry-level hiring freeze in a narrow occupational cohort before it becomes a national problem.

In April 2025, Mark Carney's Liberal platform promised to track the labour market impacts of AI. Budget 2025 backed that promise by allocating $25 million for TechStat, a new program at Statistics Canada with the mandate to measure how AI is affecting the Canadian labour force. As of April 2026, TechStat has no formal tracking framework in operation. This report proposes that the Government of Canada build an early warning and response system for AI-driven job displacement, housed within TechStat, and sets out what the system should measure, how Canada can obtain the data, and how the federal government and the provinces should prepare to act on the signals it produces. The report is written for Mila, which is well positioned to shape this system during the narrow window before TechStat's first-year work plan is finalized.

Who is at risk: Mapping the Stanford study's findings onto Canadian occupational classifications produces a canary cohort of approximately 66,705 workers aged 20 to 24 across seven occupations. Customer service representatives, software developers and engineers, and financial customer service representatives together account for roughly 78% of the cohort. Ontario hosts 43.4% of these workers (28,980), and Quebec accounts for 23.6% (15,740). This cohort represents approximately 1% of the 6 million Canadians that Statistics Canada estimates work in occupations

with high AI exposure and low complementarity. This cohort is small, but it reveals the gap between where displacement is appearing and what Canada's statistical infrastructure can detect.

What to measure: Four data streams show promise in closing this gap. AI usage statistics from frontier labs provide near-real-time visibility into how models are being used to automate or augment specific work tasks. The automation-augmentation classification developed by Anthropic has been empirically validated to correlate with employment outcomes. Dynamic task exposure metrics assess how theoretically susceptible each occupation is to AI automation as model capabilities change, and can be produced internally by TechStat with no external data agreements required. High-frequency administrative payroll data from commercial providers like ADP Canada and Dayforce shows whether displacement has reached actual workers at the occupation-by-age-by-province granularity which the LFS cannot produce. Firm-level AI adoption surveys help attribute observed employment changes to AI rather than to other economic forces.

How to get the data: Each stream carries distinct acquisition challenges. AI usage statistics are the highest-value signal but the slowest to arrive. Only Anthropic and OpenAI currently publish usage data, and the international standardization effort is unlikely to produce a formal data-sharing commitment with even a single lab before early 2027. Open-weight models (Meta's Llama, DeepSeek, and others) account for an estimated 20 to 30% of total AI usage and when users run these models on their own hardware, no usage statistics are generated. Given these constraints, TechStat should build the system on domestic data streams first and treat lab telemetry as a plug-in upgrade. Payroll data partnerships with ADP Canada and Dayforce should be structured as voluntary agreements. Statistics Canada's 2018 attempt to compel financial data under Section 13 of the Statistics Act triggered public backlash and a Privacy Commissioner investigation. The Canadian Survey on Business Conditions, Canada's only high-frequency source of firm-level AI adoption data, has its final collection cycle scheduled for August 2026. If TechStat does not absorb the AI module before it sunsets, this data stream disappears with no replacement in place.

Preparing the response: Canada's COVID-19 experience demonstrated three structural vulnerabilities that could resurface in an AI displacement scenario. ESDC's legacy benefits infrastructure, built on COBOL mainframes, could not process the surge in claims or adapt to new eligibility criteria. The government bypassed ESDC entirely and routed CERB through CRA. The inability to verify employment status in real time (CRA's most recent income data at the time of launch was from the 2018 tax year) contributed to $4.6 billion in confirmed overpayments and at

least $27.4 billion in payments flagged for further investigation. CERB payments also triggered automatic claw-backs against provincial social assistance in several provinces because there was no time to negotiate exemptions with 13 provincial and territorial social services ministries before launch. The international policy community is moving quickly on AI displacement preparedness, with the Forecasting Research Institute, Windfall Trust, and the Partnership on AI all producing scenario-based policy work in early 2026, but none of these efforts have been translated into Canada's specific institutional context: its Employment Insurance architecture, its Labour Market Transfer Agreement structure, or the federal-provincial division of responsibility for workforce adjustment.

This report proposes six recommendations organized in two groups.

RECOMMENDATIONS FOR MILA

- Brief TechStat on the indicator framework in this report and advocate for the data streams it will need, starting in Q2 2026 before TechStat's first-year work plan is finalized.
- Convene a Canadian AI labour displacement scenario exercise in October or November 2026 that produces a working document mapping plausible displacement scenarios to candidate policy responses.
- Consult provinces through bilateral outreach to Ontario and Quebec and a joint technical briefing with the Labour Market Information Council, and communicate Canada's data requirements to the international AI usage data standard effort.

RECOMMENDATIONS FOR THE GOVERNMENT OF CANADA

- TechStat should secure three domestic data streams in 2026: an LFS digital platform supplement at quarterly frequency, absorption of the CSBC AI adoption module before its August 2026 sunset, and voluntary data-sharing partnerships with ADP Canada and Dayforce.
- TechStat and ESDC's Labour Market Information Directorate should begin quarterly calibration sessions in mid-2026 to develop shared signal interpretation, escalation thresholds, and signal-to-response mapping.
- ESDC should table two sequenced items at the FLMM's Changing Nature of Work and Skills Working Group: an awareness agenda item in late 2026 and a structured working session on means-testing treatment of future AI displacement benefits in mid-2027.

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PART 1

The Canadian Landscape

CHAPTER 01

Introduction

1.1 The Imperative for Proactive Detection

The integration of generative artificial intelligence into the Canadian economy has the potential to challenge Canada's macroeconomic stability and the readiness of its social safety nets. According to Statistics Canada (StatsCan), approximately 31% of the national workforce is currently employed in occupations highly exposed to generative AI and face automation risks.[1]

Canada's labour market institutions have been built around a reactive model of detecting labour shocks through aggregate labour data, such as unemployment spikes, rather than anticipating shocks.[2] Generative AI is introducing a disruption mechanism that Canada's existing statistical apparatus is not well-equipped to detect. The earliest evidence of the labour market impacts of AI have manifested as entry-level hiring freezes in a select number of occupations which do not show up in standard government employment surveys.[3] If or when these narrow disruptions turn into broader displacement, Canada's existing response mechanisms built around reactive benefit triggers and employer-employee relationships may prove too poorly targeted to protect the workers most affected.[4] Therefore, as frontier models become increasingly capable of performing economically valuable tasks, the Government of Canada must take proactive steps towards detecting and responding to the labour impacts of AI.

[1] Tahsin Mehdi and Marc Frenette, "Exposure to Artificial Intelligence in Canadian Jobs: Experimental Estimates," Economic and Social Reports, Statistics Canada Catalogue no. 36-28-0001 (September 25, 2024), https://www150.statcan.gc.ca/n1/pub/36-28-0001/2024009/article/00004-eng.htm.
[2] Statistics Canada, "Measuring Employment and Unemployment in Canada and the United States: A Comparison," Labour Statistics: Technical Papers, Catalogue no. 75-005-M, no. 2015002 (Ottawa: Statistics Canada, 2015), https://www150.statcan.gc.ca/n1/pub/75-005-m/75-005-m2015002-eng.htm.
[3] Tahsin Mehdi and Marc Frenette, "Canadian Employment Trends in the Era of Generative Artificial Intelligence: Early Evidence," Economic and Social Reports, Statistics Canada Catalogue no. 36-28-0001 (January 28, 2026), https://www150.statcan.gc.ca/n1/pub/36-28-0001/2026001/article/00003-eng.htm.
[4] Employment and Social Development Canada, "EI Regular Benefits: Do You Qualify," Government of Canada, accessed April 7, 2026, https://www.canada.ca/en/services/benefits/ei/ei-regular-benefit/eligibility.html.

1.2 The Policy Window

In April 2025, Mark Carney's Liberal campaign platform promised to "track labour market impacts of AI to rapidly adjust federal training programs," a commitment assigned to Minister of Jobs and Families Patty Hajdu with support from Minister of Artificial Intelligence and Digital Innovation.[5] Canada's Budget 2025 also announced TechStat, a new initiative housed within Statistics Canada to track the adoption and impact of AI on the Canadian economy.[6] As of March 2026, an independent government accountability tracker assessed that while preliminary steps have been taken towards this commitment, no formal tracking framework or dashboard monitoring AI impacts on Canadian occupations has been confirmed as operational.[7] The system proposed in this report is designed to fill this gap. This domestic mandate intersects with a parallel international effort to develop standardized AI usage reporting from frontier labs, discussed in Chapter 6. In this context, Canada has an opportunity to shape both the domestic monitoring infrastructure and the international data standards it will eventually rely on. Taking concrete steps on this front, in combination with Canada's forthcoming National AI Strategy, can establish Canada as a leader in responsible AI adoption.[8]

1.3 The Central Question

This report addresses the following policy question: ***How can Canada build an early warning and response system for AI-driven job loss?*** Answering this question requires knowing which specific indicators can detect a localized labour shock before it appears in national statistics, how Canada can actually obtain those indicators, and what the government must do when the warning signals start flashing.

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[5] Liberal Party of Canada, Canada Strong: Unite. Secure. Protect. Build. (2025 federal election platform), April 2025, https://liberal.ca/wp-content/uploads/sites/292/2025/04/Canada-Strong.pdf; Build Canada, "Track AI Labor Market Impacts to Rapidly Adjust Federal Training Programs," Build Canada Outcomes Tracker, accessed April 7, 2026, https://www.buildcanada.com/tracker/commitments/2524.

[6] Department of Finance Canada, "Chapter 1: Building a Stronger Canadian," in Budget 2025: Canada Strong (Ottawa: Government of Canada, November 4, 2025), https://budget.canada.ca/2025/report-rapport/chap1-en.html.

[7] Build Canada, "Track AI Labor Market Impacts to Rapidly Adjust Federal Training Programs," Build Canada Outcomes Tracker, accessed April 7, 2026.

[8] Baker McKenzie, "Canada: Insights from the National AI Strategy Consultation," Baker McKenzie Insight, March 2026, https://www.bakermckenzie.com/en/insight/publications/2026/03/canada-insights-from-national-ai-strategy-consultation.

CHAPTER 02

Report Overview and Methodology

2.1 Purpose of this Report

Recent research has identified a promising set of indicators that may be capable of providing early signals for AI-driven labour market disruptions. However, no government has yet built an operational system that uses these indicators for ongoing monitoring and policy response.

This report provides Mila with a roadmap for helping the Government of Canada operationalize TechStat and turn it into a functioning early warning system. It does this by identifying the specific indicators TechStat should be tracking, mapping the data partnerships and legal authorities Canada can use to access them, assessing which components are available immediately and which will take longer to build, and identifying the federal and provincial stakeholders Mila must engage to ensure the system produces trusted signals that lead to effective policy responses. This report does not propose specific benefit designs, funding levels, or legislative changes.

While Mila is the primary audience, this report is also designed to serve as a tool Mila can bring into its engagements with government stakeholders. Federal and provincial officials reading this document should come away with a clear understanding of what the early warning system measures, why it matters, and what their institution's role is in making it work.

To support these engagements, the appendices provide methodological detail, legal and procedural guidance for the proposed data partnerships, supplementary data breakdowns, and a reference table of the key decision-makers relevant to each recommendation.

2.2 Structure of the Report

To achieve this purpose, the analysis unfolds in three parts:

- **Part 1 - The Canadian Context (Chapter 3):** Documents the divergence between macro-level employment stability and micro-level disruption among specific groups of younger workers, quantifies the Canadian population most immediately at risk, and identifies the vulnerabilities in Canada's existing safety net and statistical infrastructure.

- **Part 2 - Building the System (Chapters 4-7):** Introduces the roadmap for building the early warning system and the key stakeholders involved (Chapter 4), then addresses each step in the roadmap in sequence: what should Canada be measuring? (Chapter 5), how does Canada get access to that data? (Chapter 6), and how should the government prepare to act on the data when it arrives? (Chapter 7).
- **Part 3 - The Conclusion (Chapters 8):** Consolidates the action items shared throughout the report into six prioritized recommendations (three for Mila and three for the Government of Canada) and provides an implementation plan with specific timelines, stakeholders, and venues.

Readers primarily interested in the data and measurement architecture should focus on Chapters 5 and 6. Readers focused on policy response design and federal-provincial coordination should read Chapter 7. Readers looking for specific action items and timelines should go directly to Chapter 8 and the implementation timeline in Section 8.4. **Appendix F** provides a reference table of the individual decision-makers and counterparts whose authority is relevant to each recommendation.

2.3 Methodology

This report used a mixed-methods approach in three phases:

Phase 1 - Literature Review & Indicator Analysis: The first phase consisted of a targeted review of the latest research on generative AI's workforce impacts, including theoretical AI exposure metrics, emerging evidence regarding AI-usage statistics (such as the Anthropic Economic Index), high-frequency payroll data, and employment trends from Statistics Canada. The objective was to identify which indicators are most promising for detecting labour disruptions, and analyze the current state of Canada's workforce and safety nets.

Phase 2 - Qualitative Expert Consultations: The second phase utilized semi-structured interviews to understand the institutional and political realities of the Canadian government, and identify international AI governance efforts that could be leveraged by Canada to access AI usage statistics from the AI labs.

On the domestic side, the report draws on interviews with two former federal insiders:

- **Tyler Meredith**, former Lead Economic Advisor to Prime Minister Justin Trudeau, who provided guidance on navigating Statistics Canada's institutional dynamics, including the central role of the Chief Statistician in approving changes to survey instruments, the role of ISED (the ministry Statistics Canada reports to) and ESDC as potential channels for advancing TechStat's work, and the limitations of existing surveys like the LFS for detecting narrow occupational disruptions.
- **Jaxson Khan**, former Senior Policy Advisor to Canada's Minister of Innovation, Science, and Industry, who shared perspectives from his role in designing Canada's Sovereign AI Compute Strategy and offered guidance on how to build institutional credibility for new data initiatives within the federal government.

On the international side, interviews were conducted with **Sam Manning**, a Senior Research Fellow at the Centre for the Governance of AI (GovAI), and **Ketana Krishna**, a contractor with the Oxford Martin AI Governance Initiative. They provided insights from their work coordinating a multi-country coalition to develop standardized reporting requirements for AI usage statistics and create voluntary data-sharing agreements with frontier AI labs.

Phase 3 - Historical Policy Case Study: The third phase examined Canada's COVID-19 economic response as a case study in emergency benefit delivery under time pressure. The analysis identified specific structural vulnerabilities in Canada's benefits infrastructure, eligibility verification systems, and federal-provincial coordination that may resurface in an AI displacement scenario. These findings informed the report's recommendations.

CHAPTER 03

The Canadian Landscape: Threats and Vulnerabilities

CHAPTER AT A GLANCE

Approximately 6 million Canadians work in occupations theoretically exposed to AI, but early empirical evidence of displacement has appeared in only about 1% of that population: roughly 66,705 early-career workers aged 20 to 24 across seven specific occupations. The largest groups within this canary cohort are customer service representatives (29,605) and software developers and engineers (13,810 combined). Ontario and Quebec together hold 67% of the cohort, with Ontario alone accounting for 43.4% (28,980 workers). Three findings from this analysis shape the design of the early warning system proposed in this report: theoretical exposure metrics are too broad to guide monitoring on their own, the canary cohort is too small and too localized to be detected by Canada's existing monthly survey instruments, and Canada's primary safety net (Employment Insurance) is built on the premise of prior employment, which many workers in this early-career cohort would not have accumulated.

3.1 Macro-Stability vs. Micro-Disruption

As of early 2026, assessments of AI's labour market impact tell two diverging stories depending on the granularity of the analysis. At the aggregate level, Canadian employment has been stable and growing regardless of jobs' exposure to AI. A January 2026 Statistics Canada report examining employment trends from November 2022 to December 2025 found that jobs classified as high exposure and low complementarity (HELC) under Statistics Canada's complementarity-adjusted AI occupational exposure index (C-AIOE) grew by approximately 10% for men and 5% for women.[9] According to a 2024 assessment using the same index, approximately 31% of the Canadian workforce, roughly 6 million people, were employed in HELC occupations. At this level of resolution, AI appears to be having little measurable effect on Canadian employment.[10]

[9] Frenette, "Canadian Employment Trends."
[10] Mehdi and Frenette, "Experimental Estimates of Potential Artificial Intelligence Occupational Exposure in Canada."

Beneath this aggregate stability, the same report also found disruptions within particular groups of younger workers. In highly exposed, coding-intensive professions, employment for 15- to 29-year-old Canadians remained stagnant between late 2022 and late 2025, while employment for 30- to 49-year-olds in the same field grew by almost 30%. Canadian job vacancies for coding roles requiring three years of experience or less plummeted by about 60% over a similar period.[11]

This divergence between macro-level resilience and micro-level disruption is mirrored in the United States. Recent US assessments by the Budget Lab at Yale found that on a national level, AI exposure does not correlate with rising unemployment.[12] However, when the focus is narrowed to specific occupations (e.g., software developers, customer service representatives) the 'Canaries in the Coal Mine' paper by Brynjolfsson et al. found a 16% relative drop in employment for young workers aged 22-25 in those roles compared to their peers since 2022.[13]

Brynjolfsson et al. explicitly controlled for confounding variables, including firm-level shocks, post-pandemic labour adjustments, and the fallout from the 2021-2022 tech hiring boom.[14] Conversely, the StatsCan report observed a similar correlational trend but noted their data could be confounded by these exact variables, alongside demographic shifts from immigration.[15] Thus, while Canadian policymakers must interpret these results with caution, they suggest that generative AI may pose a real threat to entry-level workers in specific sectors through reduced hiring.

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[11] Ibid.
[12] Martha Gimbel, Molly Kinder, Joshua Kendall, and Maddie Lee, "Evaluating the Impact of AI on the Labor Market: Current State of Affairs," The Budget Lab at Yale, October 1, 2025, https://budgetlab.yale.edu/research/evaluating-impact-ai-labor-market-current-state-affairs.
[13] Erik Brynjolfsson, Bharat Chandar, and Ruyu Chen, "Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence" (working paper, Stanford Digital Economy Lab, November 2025), https://digitaleconomy.stanford.edu/app/uploads/2025/11/CanariesintheCoalMine_Nov25.pdf.
[14] Ibid.
[15] Frenette, "Canadian Employment Trends."

3.2 Quantifying Canada's At-Risk Working Population

While 6 million workers were estimated to be employed in HELC occupations, adoption frictions, workflow reorganization costs, and uneven AI deployment across firms mean that theoretical exposure alone is a poor predictor of which workers will actually experience displacement.[16]

To identify the Canadian cohorts most at risk of AI-driven job displacement, this report draws on the Brynjolfsson et al. framework introduced in the Section 3.1. In addition to theoretical exposure scores, Brynjolfsson et al. also used real-world evidence from AI usage statistics. The 16% relative employment decline they found appeared only in occupations that scored high on both theoretical exposure and automation-oriented usage (i.e. where AI models were being used in an automation-oriented capacity for tasks associated with those occupations).[17]

The U.S. occupational codes in which Brynjolfsson et al. found the employment decline were mapped to their closest equivalents in Statistics Canada's National Occupational Classification system by cross-referencing SOC-to-NOC correspondence tables.[18] Five U.S. occupations (software developers, customer service representatives, bank tellers, travel agents, and tax preparers) yield seven Canadian NOC 2021 codes, as illustrated in Figure 1.

This approach relies on the following logic: If the U.S. analysis found specific occupations and cohorts where labour displacement is correlated with generative AI after controlling for confounding variables, then mapping those occupations to their Canadian equivalents is a defensible starting point within the scope of this report for identifying Canadian workers most likely to be affected by similar dynamics. However, the U.S. and Canadian labour markets are not identical, and this mapping does not guarantee that the same pattern will manifest in Canada.

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[16] Mehdi and Frenette, "Exposure to Artificial Intelligence in Canadian Jobs;" Epoch AI, "Challenges in Predicting AI Automation," Epoch AI (blog), November 24, 2023, https://epoch.ai/blog/challenges-in-predicting-ai-automation.

[17] Brynjolfsson, Chandar, and Chen, "Canaries in the Coal Mine?"

[18] Statistics Canada, "Concordance between the 2018 U.S. Standard Occupational Classification (SOC) and the 2016 National Occupational Classification (NOC) Version 1.3," Government of Canada, accessed April 7, 2026, https://www.statcan.gc.ca/en/statistical-programs/document/soc2018US-noc2016v1_3; Statistics Canada, " Correspondence Table: National Occupational Classification (NOC) 2016 V1.3 to National Occupational Classification (NOC) 2021 V1.0 based on GSIM," Government of Canada, accessed April 7, 2026, https://www.statcan.gc.ca/en/statistical-programs/document/noc2016v1_3-noc2021v1_0.

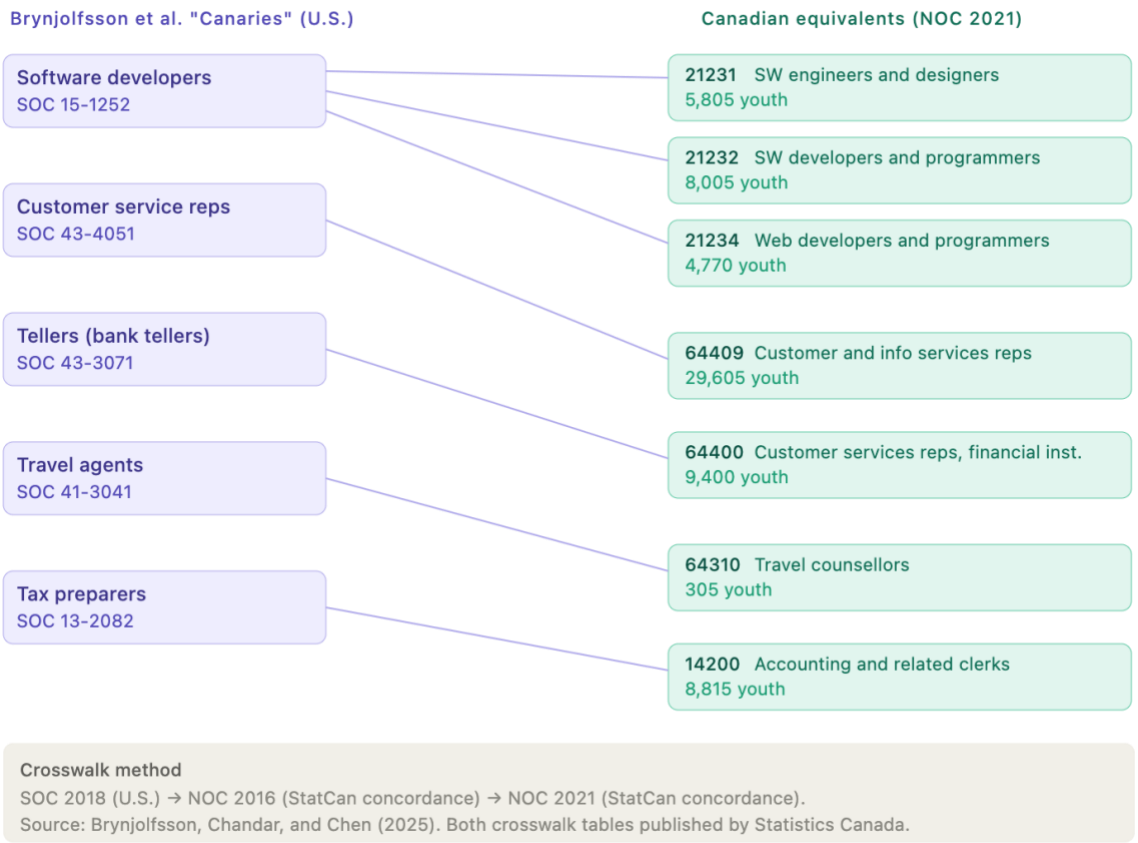
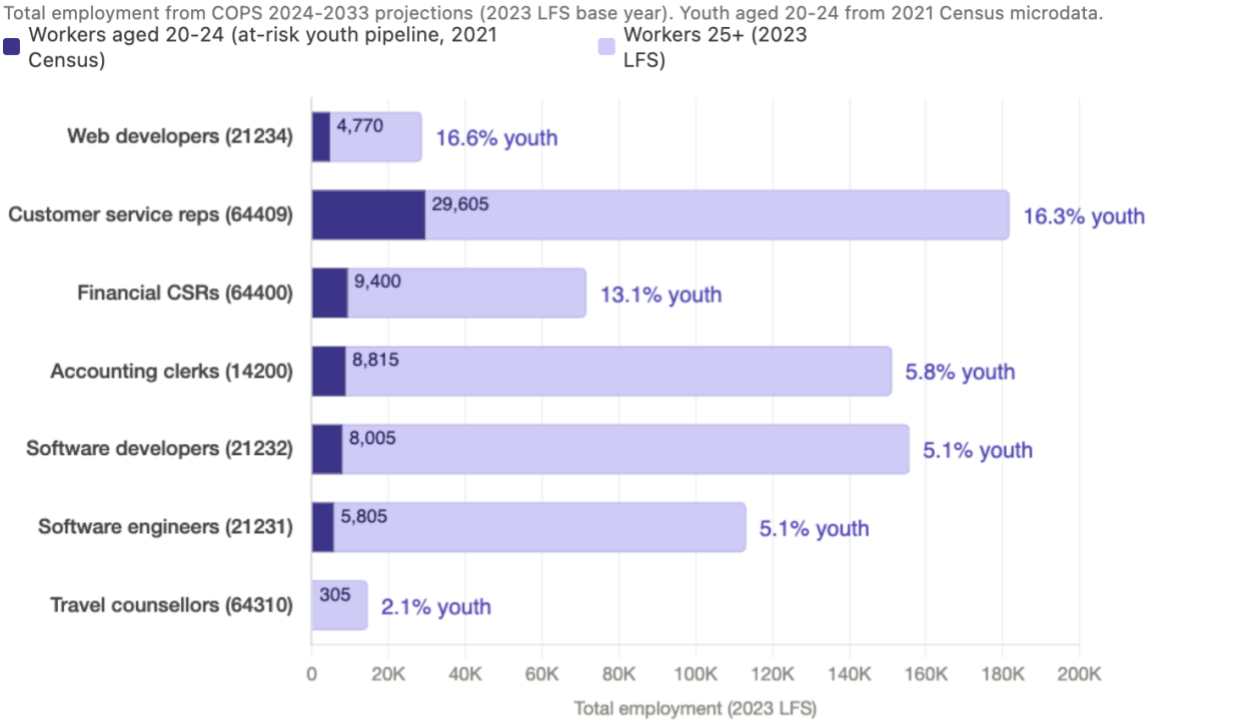


Figure 1. Crosswalk from U.S. to Canadian occupational classifications: identifying Canada's canary occupations

Note: The five U.S. occupations in which Brynjolfsson, Chandar, and Chen (2025) found early-career employment declines correlated with generative AI adoption were mapped to Canadian equivalents using two Statistics Canada correspondence tables: SOC 2018 to NOC 2016, and NOC 2016 to NOC 2021. Software Developers (SOC 15-1252) maps to three Canadian codes due to differences in how the two classification systems subdivide software occupations. Youth counts are employed persons aged 20 to 24 from Statistics Canada Table 98-10-0593-01 (2021 Census, 25% sample).

Applying the early-career demographic filter (workers aged 20 to 24, the closest available Canadian proxy) to those seven occupations using 2021 Census microdata produces a canary cohort of approximately **66,705 young workers**. These are the workers this report refers to as the "canary cohort": the 1% of the ~6 million theoretically exposed Canadians who sit at the intersection of high-exposure occupations, automation-oriented AI usage, and early-career age.

Figure 2 shows the national breakdown by occupation. The three largest groups by volume are customer service representatives (29,605), software developers and engineers (13,810 combined), and financial customer service representatives (9,400), together accounting for roughly 78% of the cohort.



66,705

early-career workers aged 20 to 24 across seven high-exposure occupations form Canada's canary cohort for AI displacement.

43.4%

concentrated in Ontario (28,980 workers). Ontario and Quebec together hold 67% of the cohort.

Figure 2. Canada's "Canary" Occupations: Total Employment vs. At-Risk Youth Pipeline

3.2.1 Geographic Concentration

The canary cohort is not evenly distributed across Canada. Using the same 2021 Census microdata used to calculate the national cohort (Table 98-10-0593-01), this report extracted employment counts for the seven canary occupations at the provincial level, filtered to workers aged 20 to 24.[19]

As Figure 3 illustrates, the cohort is heavily concentrated in Ontario and Quebec. Ontario accounts for 43.4% of the canary cohort (28,980 workers), and Quebec accounts for 23.6% (15,740 workers). Together, these two provinces hold two-thirds of Canada's at-risk early-career workers in AI-

[19] Statistics Canada, "Class of Worker by Occupation Unit Group, Labour Force Status, Age and Gender: Canada, Provinces and Territories, Census Metropolitan Areas and Census Agglomerations with Parts," Table 98-10-0593-01, 2021 Census of Population, https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=9810059301.

exposed occupations. British Columbia (12.5%), Alberta (7.0%), and Manitoba (5.1%) round out the top five.

Source: Statistics Canada, Table 98-10-0593-01, 2021 Census of Population (25% sample). Employed persons aged 20-24 in seven canary NOC codes. Provincial totals sum to 66,695 due to rounding.

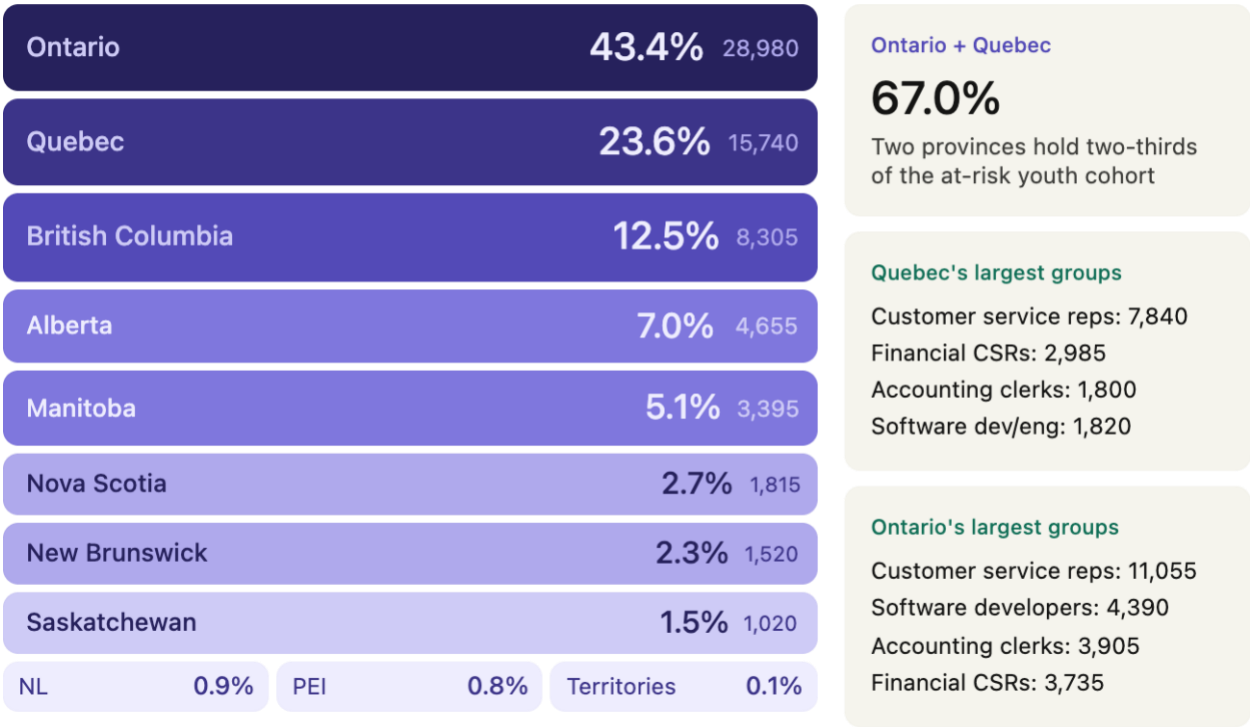
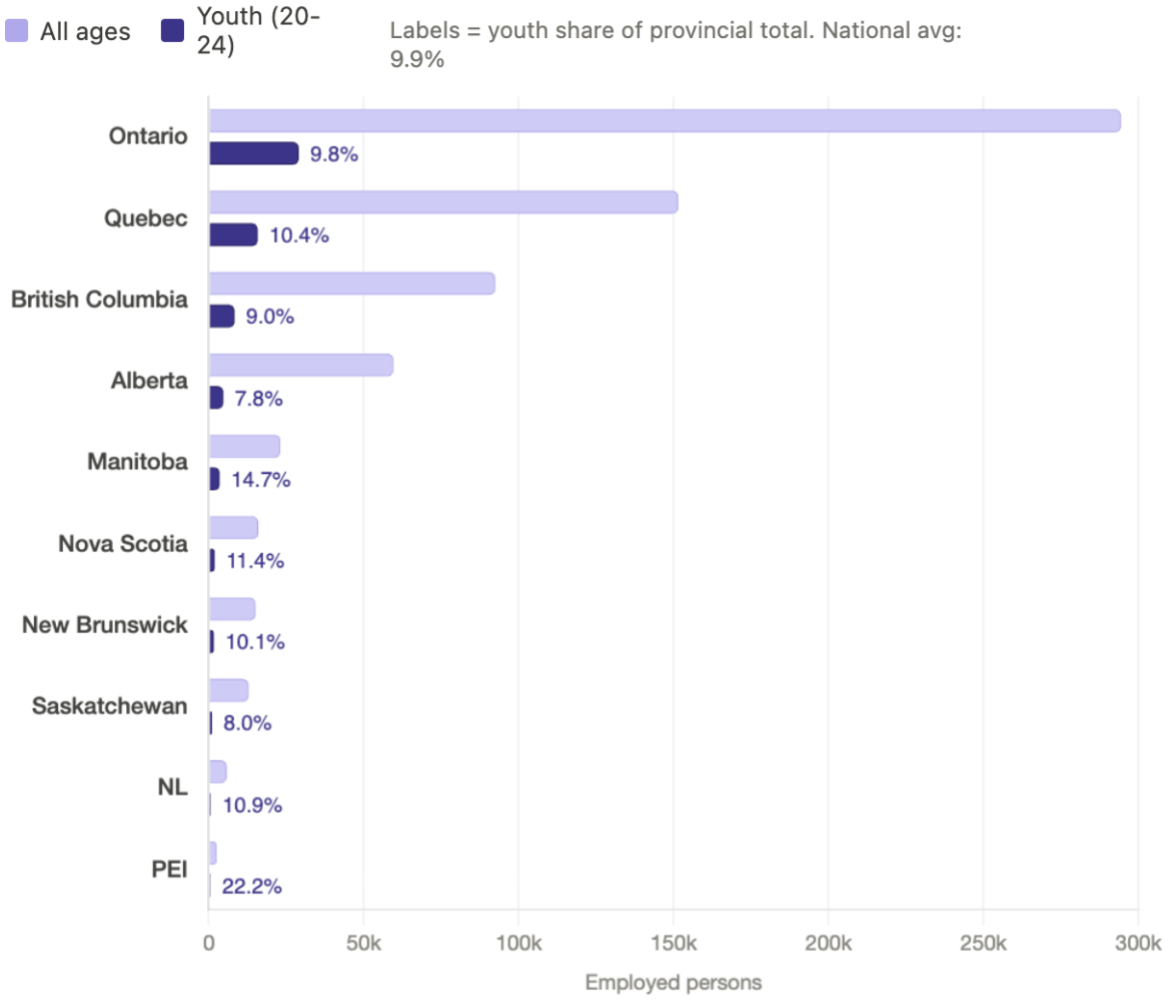


Figure 3. Provincial Distribution of Canada's Youth Canary Cohort (Workers Aged 20-24)

Note: Figure 3 reflects employed persons aged 20 to 24 in the seven canary occupations, extracted from Statistics Canada Table 98-10-0593-01 (2021 Census, 25% sample, reference week of May 2-8, 2021). Provincial totals sum to 66,695 due to StatCan rounding of individual cells to the nearest 5.

Ontario's canary cohort is the largest in absolute terms, but its youth share of 9.8% is close to the national average of 9.9% (Figure 4). Ontario's dominance reflects the province's overall size and the concentration of financial services headquarters and technology employers in Toronto and Ottawa.



Source: Statistics Canada, Table 98-10-0593-01, 2021 Census of Population (25% sample). Seven canary NOC codes (14200, 21231, 21232, 21234, 64310, 64400, 64409).

Figure 4. Total employment vs. youth pipeline in canary occupations by province

Ontario has the largest workforce in the seven exposed occupations (294,335 workers across all ages, of whom 28,980 are in the 20-to-24 canary cohort), but its youth share of 9.8% is close to the national average of 9.9%. Ontario's dominance of the canary cohort reflects the province's overall size and the concentration of financial services headquarters and technology employers in Toronto and Ottawa. The policy implication is one of scale: even at an average youth share, Ontario's absolute count of nearly 29,000 canary workers means that a displacement signal would be felt most acutely in Ontario simply because of the number of people involved.

The seven occupations are distributed differently across provinces. A full provincial breakdown by occupation is provided in **Appendix E**: Canary Cohort Occupational Composition by Province.

3.3 Interpreting the Canary Cohort

The preceding analysis narrows the population of concern from 6 million theoretically exposed Canadians to approximately 67,000 workers who sit at the intersection of confirmed high-exposure occupations, automation-oriented AI usage, and early-career age. Analyzing this cohort leads to three key insights that must shape the design of Canada's early warning system.

First, theoretical exposure alone is too broad to be a meaningful metric for monitoring and policymakers must be cautious to not over-weight this metric. The approximately 67,000 workers in the canary cohort represent only 1% of the 6 million Canadians in theoretically exposed (HELC) occupations. Whether and when displacement spreads beyond this 1% depends on variables like how AI tools are being used to complete specific tasks, how quickly firms adopt and integrate them, and whether AI capabilities advance into new task domains.

Second, the canary cohort falls below the detection threshold of standard government survey instruments. The 'Canaries in the Coal Mine' researchers found that the U.S. Current Population Survey (the American equivalent of Canada's Labour Force Survey) captured only between 26 and 53 young software developers per month in their sample. At that scale, a genuine hiring freeze would be indistinguishable from normal sampling noise.[20] Canada's Labour Force Survey (LFS) captures self-employment status, broad occupational categories, and class of worker on a monthly basis. It is the only source of monthly estimates for the entire labour force, and is the most representative labour market instrument Statistics Canada operates. However, LFS faces the same constraint with a sample size of approximately 65,000 households.[21] A hiring freeze affecting customer service representatives aged 20 to 24 in Manitoba (2,220 workers) would be invisible to the LFS. Chapter 5 explains how payroll data can close this detection gap.

For a breakdown of the limitations of specific Canadian instruments like the LFS, and SEPH please refer to **Appendix B**: The Limits of the Canadian Statistical Apparatus.

Third, Canada's safety nets have multiple vulnerabilities that directly affect the canary population. The most immediate vulnerability for the canary cohort is Employment Insurance (EI)

[20] Brynjolfsson, Chandar, and Chen, "Canaries in the Coal Mine?"
[21] Statistics Canada, "Labour Force Survey (LFS)," Surveys and Statistical Programs, accessed April 7, 2026, https://www23.statcan.gc.ca/imdb/p2SV.pl?Function=getSurvey&SDDS=3701.

eligibility. Canada's EI system is built on the premise of prior employment.[22] Many workers aged 20 to 24 would be entering the labour market for the first time and would fall outside EI eligibility when faced with an entry-level hiring freeze.

The eligibility gap also extends into the Labour Market Development Agreements (LMDAs), which is the primary channel for provinces to receive funding from the federal government for retraining programs. LMDAs transfer approximately $2 billion annually from EI premium revenues to provinces for skills programming, but are restricted to current or former EI claimants.[23] Workforce Development Agreements (WDAs), which is a secondary channel funded at $722 million annually from general revenues, can theoretically reach canary workers because they serve individuals regardless of EI attachment.[24] However, the twelve provinces and territories that took part in ESDC's 2022 evaluation of the WDA program all identified insufficient funding as their most pressing challenge in a 2022. This concern came after Budget 2017 temporarily increased both LMDA and WDA transfers.[25] When those top-ups expired in 2023-24, it introduced a cut of roughly $625 million annually to LMDAs and WDAs collectively, and triggered a provincial boycott of the federal Workforce Summit in 2024.[26] The financial capacity of WDAs and eligibility requirements of LMDAs create constraints for any retraining initiatives required in response to AI-driven displacements. Based on the COVID-19 precedent, Canada's likely emergency fallback option would be a CRA-delivered benefit. The CRA can set up a cash transfer within weeks but that channel would only activate once the impact is highly visible and politically viable. Canada's COVID-19 response also revealed significant costs to deploying emergency benefits without the

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[22] Employment and Social Development Canada, "Employment Insurance (EI)," Government of Canada, accessed April 12, 2026, https://www.canada.ca/en/employment-social-development/programs/ei.html.
[23] Employment and Social Development Canada, "Labour Market Development Agreements," Government of Canada, accessed April 12, 2026, https://www.canada.ca/en/employment-social-development/programs/training-agreements/lmda.html.
[24] Employment and Social Development Canada, "Workforce Development Agreements," Government of Canada, accessed April 7, 2026, https://www.canada.ca/en/employment-social-development/programs/training-agreements/workforce-development-agreements.html.
[25] Employment and Social Development Canada, Evaluation of the Workforce Development Agreements (Ottawa: Government of Canada), accessed April 12, 2026, https://www.canada.ca/en/employment-social-development/corporate/reports/evaluations/workforce-development-agreements.html.
[26] Forum of Labour Market Ministers, "Provincial and Territorial Labour Market Ministers Deeply Concerned about Federal Government's Approach to Workforce Summit 2024" (news release, Cision Newswire, October 16, 2024), https://www.newswire.ca/news-releases/provincial-and-territorial-labour-market-ministers-deeply-concerned-about-federal-government-s-approach-to-workforce-summit-2024-827266061.html; Employment and Social Development Canada, "Labour Market Transfer Agreements," Government of Canada, accessed April 12, 2026, https://www.canada.ca/en/employment-social-development/programs/training-agreements.html.

infrastructure to verify employment status in real-time.[27] This case study will be examined in Chapter 7. These insights highlight vulnerabilities in theoretical exposure metrics, Canada's survey instruments, and safety nets. Some of these are particularly relevant to the early-career canary cohort (e.g. EI eligibility), but note that the early warning system this report proposes cannot be designed solely for this population. Disruption to entry level software developers and customer service representatives is a pattern that has been observed and requires attention, but an effective monitoring system must be built to detect displacement signals across the labour market.

Chapter 4 introduces the building blocks of that system: the questions it must answer, and the stakeholders required to build and operate it.

[27] Office of the Auditor General of Canada, "Report 10: Specific COVID-19 Benefits," in 2022 Reports of the Auditor General of Canada to the Parliament of Canada (Ottawa: Office of the Auditor General, December 2022), https://www.canada.ca/en/auditor-general/our-work/audit-reports/parl-oag-202212-10-e.html.

CHAPTER 04

Building the Early Warning System

Building the system this report proposes requires answering three questions:

1. **What should Canada be measuring?** The answer involves identifying the specific data streams that can reveal AI-driven displacement before it surfaces in national statistics.
2. **How can Canada get access to the right data?** Some of it is available immediately through domestic survey instruments, and some of it depends on voluntary data partnerships and international coordination that will take at least a year.
3. **How should the government prepare to act on the data once it arrives?** This includes building confidence in the signals by demonstrating their statistical significance, gaining buy-in from the provinces and ESDC so they are ready to act based on the signal, and developing a pre-planned response protocol to early warning signals.

Chapters 5, 6, and 7 address these three questions in sequence. Before turning to them, this chapter identifies the key stakeholders involved across all three.

4.1 Key Stakeholders

TechStat: Formally the 'Artificial Intelligence and Technology Measurement Program,' this program sits within Statistics Canada and is the natural home for the early warning system. TechStat is funded through Budget 2025 ($25 million over six years), and has the mandate to use data and insights to understand AI's impact on the labour force.[28] It must define what data the government needs, secure it from relevant data partners, and validate whether the resulting signals are reliable enough to support policy decisions. TechStat is new and its work plan is still taking shape, which means Mila has a window to influence what TechStat prioritizes and how it builds the early warning system.

Employment and Social Development Canada (ESDC): ESDC is the policy response owner. Once TechStat identifies a credible labour shock, ESDC must activate the appropriate intervention based

[28] Department of Finance Canada, "Chapter 1," Budget 2025: Canada Strong.

on the severity of the expected shock whether through Employment Insurance adjustments, Labour Market Transfer Agreements (LMTA), or emergency benefit delivery. Within ESDC, the Labour Market Information Directorate, nested within the Skills and Employment Branch, is the unit most relevant to the early warning system.[29] The Directorate produces and manages labour market intelligence for policy use, and coordinates with Statistics Canada on survey products. It is the natural internal consumer of TechStat's outputs and the unit that would interpret early warning signals and advise on whether to escalate to a policy response. Paul Thompson has served as Deputy Minister of Employment and Social Development since September 2023 and oversees the department's full organizational structure, including the Skills and Employment Branch where the Labour Market Information Directorate sits.[30]

The Centre for the Governance of AI (GovAI) and The Oxford Martin AI Governance Initiative: A team led by Sam Manning at GovAI is developing a proposed industry standard for how frontier AI labs should report anonymized usage statistics. The effort involves defining the exact reporting format, including how usage should be mapped to occupational task taxonomies and at what level of geographic and industry detail the data should be disaggregated. Manning's team have been engaging government contacts across several countries, including Canada, India, Singapore, Switzerland, and Brazil, to build support for a coordinated request to labs. The India AI Impact Summit in February 2026 provided a platform for advancing this agenda, which resulted in general commitments from participating governments.[31] Mila's role is to ensure the emerging standard meets Canada's specific needs (discussed in Chapter 6) and to prepare federal and provincial officials to use this data when it arrives.

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[29] Employment and Social Development Canada, "About Us," Government of Canada, accessed April 12, 2026, https://www.canada.ca/en/employment-social-development/corporate.html; Labour Market Information Council, "LMI Insight Report No. 35: Job-Skills Mapping: Building a Concordance between the U.S. O*NET System and Canada's NOC" (report, LMIC-CIMT, Ottawa, September 2020), https://lmic-cimt.ca/publications-all/lmi-insight-report-no-35-job-skills-mapping-building-concordance-between-the-us-onet-system-and-canadas-noc/.

[30] Government of Canada, "Paul Thompson, Deputy Minister of Employment and Social Development Canada," Canada.ca, accessed April 2026, https://www.canada.ca/en/employment-social-development/corporate/official-paul-thompson.html.

[31] [31] Press Information Bureau, Government of India, "Championing Inclusive and Multilingual AI for the Global South, India Unveils New Delhi Frontier AI Commitments" (press release), https://www.pib.gov.in/PressReleasePage.aspx?PRID=2230201&lang=2&reg=3.

The Provinces and Territories: While the federal government provides funding through LMTAs, provincial governments design and deliver the actual retraining and adjustment programs.[32] As the geographic concentration data in Chapter 3 illustrated, Ontario is likely to be disproportionately impacted by a canary-style junior hiring freeze. Buy-in from the provinces and territories is critical to the success of the early warning system, and Mila's provincial engagement should account for their needs, existing infrastructure and capabilities.

The Forum of Labour Market Ministers (FLMM): This forum has convened labour market ministers from all 13 provinces and territories, ESDC, and Statistics Canada since 1983. It is currently co-chaired by Minister Hajdu and Nova Scotia's Minister of Labour, Skills and Immigration. There are four working groups, including the Changing Nature of Work and Skills Working Group, which is the natural venue to raise an AI labour displacement agenda item.[33] FLMM is the right primary convening table for Mila and TechStat to engage provincial labour ministers.

The implementation plan for Recommendation 3 in Chapter 8 details Mila's provincial engagement strategy, and Recommendation 6 addresses ESDC's provincial engagement on benefit coordination.

Mila: Mila's role is to ensure that the relevant government bodies are well-informed on the latest research regarding useful indicators, understand the most effective pathways to securing the required data, and are ready to work together so that TechStat does not operate in a vacuum.

Appendix F provides a reference table of the individual decision-makers and counterparts within each institution whose authority is most relevant to this report's recommendations.

[32] ESDC, "Labour Market Agreements."
[33] Forum of Labour Market Ministers, "Changing Nature of Work and Skills," FLMM-FMMT, accessed April 12, 2026, https://flmm-fmmt.ca/initiatives/changing-nature-of-work-and-skills/.

PART 2

Building the System

CHAPTER 05

What Canada Needs to Measure

CHAPTER AT A GLANCE

The early warning system requires four data streams, each with distinct strengths and acquisition challenges.

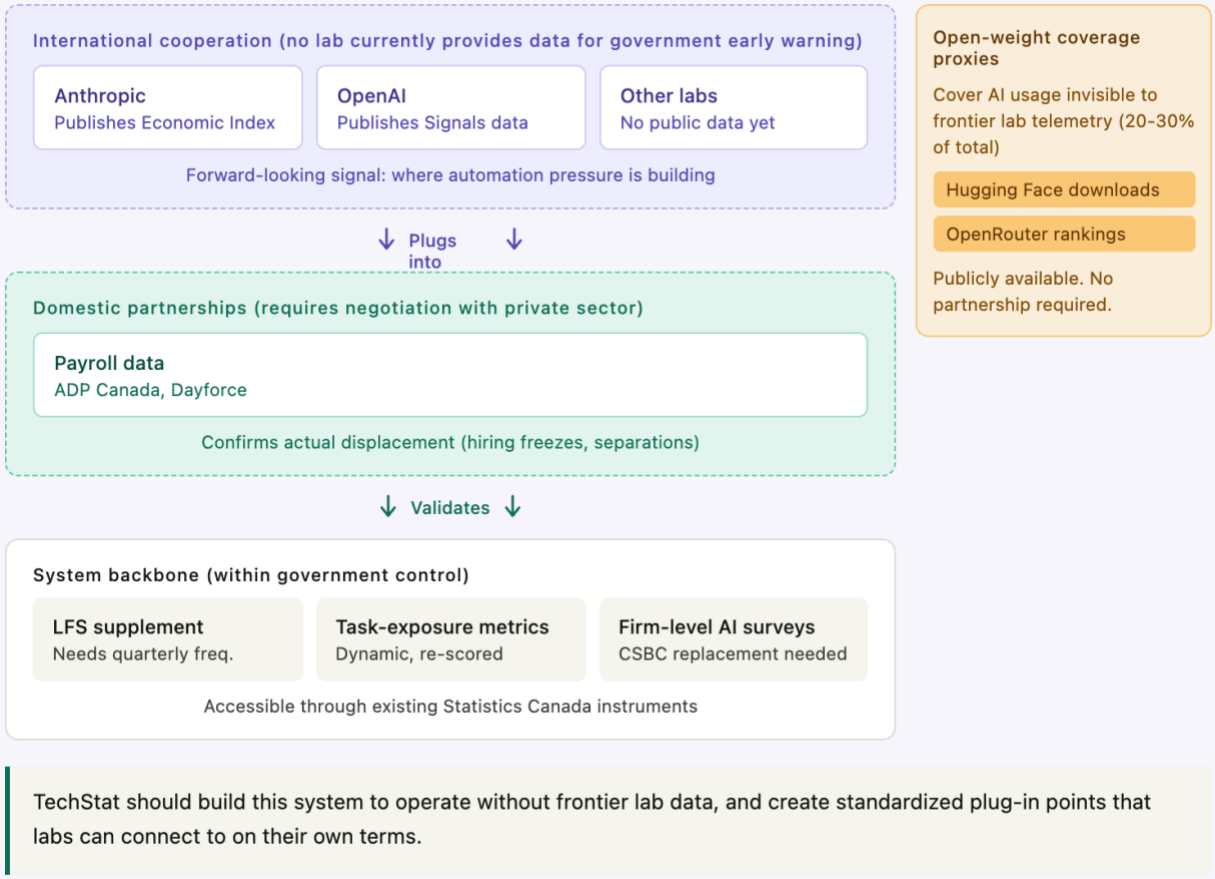
***AI usage statistics** offer near-real-time visibility into how people use AI to complete work tasks. The automation-augmentation classification has been empirically validated as a predictor of employment outcomes, making it a promising signal. Anthropic and OpenAI are the only labs currently publishing usage data, and formal data-sharing agreements will take at least a year to secure. Multi-lab coverage matters because usage patterns differ sharply across platforms: roughly one-third of Claude interactions involve coding, compared with approximately 4% of ChatGPT messages. Open-weight models (Meta's Llama, DeepSeek, Qwen, and others) account for an estimated 20 to 30% of total AI usage. AI usage statistics are a high-value data stream, but TechStat should build the system to function without them and treat lab data as a plug-in upgrade.*

Dynamic task exposure metrics assess how susceptible each occupation is to AI automation as model capabilities change. They can be produced internally by TechStat using frontier models to score the full National Occupational Classification, with no data-sharing agreements or administrative data required. Static exposure indices like Statistics Canada's existing C-AIOE are anchored in pre-generative-AI methodology and become outdated as capabilities advance, but dynamic re-scoring addresses this.

High-frequency payroll data confirms whether displacement has reached actual workers, providing the near-real-time hiring and separation signals that the LFS cannot produce at the occupation-by-age-by-province granularity the early warning system needs. Commercial providers cover an estimated 80% of the Canadian workforce. Acquiring this data requires voluntary partnerships with ADP Canada and Dayforce, which Chapter 6 addresses.

Firm-level AI adoption surveys help attribute observed employment changes to AI rather than to other economic forces. The Canadian Survey on Business Conditions (CSBC) has carried AI adoption questions since Q2 2024, but its final collection cycle is scheduled for summer 2026. TechStat needs a

replacement survey vehicle before it loses Canada's only high-frequency source of firm-level AI adoption data.



TechStat should build this system to operate without frontier lab data, and create standardized plug-in points that labs can connect to on their own terms.

Figure 5. Early warning system architecture: data streams by level of dependency

5.1 AI Usage Statistics

AI usage statistics are anonymized, aggregated statistics that frontier AI labs produce by classifying how users interact with their models.[34] As discussed in Chapter 3, the automation-augmentation classification developed by Anthropic is currently the only AI usage metric that has been empirically validated against real-world employment outcomes through the Canaries in the Coal Mine paper.[35] Anthropic's research program has also produced additional metrics including task complexity (how long a task takes to complete without AI), human education requirements (years

[34] Ruth Appel, Maxim Massenkoff, Peter McCrory, Miles McCain, Ryan Heller, Tyler Neylon, and Alex Tamkin, "Anthropic Economic Index Report: Economic Primitives" (Anthropic, January 15, 2026), https://www.anthropic.com/research/anthropic-economic-index-january-2026-report.
[35] Brynjolfsson, Chandar, and Chen, "Canaries in the Coal Mine?"

of education needed to perform the task), AI autonomy levels (how much the user delegates decision-making to the model), and task success rates (whether the model actually completes the task effectively).[36] Each of these additional metrics may carry policy implications as the evidence base grows. This points to a broader need for further empirical validation of AI usage metrics in the Canadian workforce.

5.1.1 Which Labs Publish Data

Anthropic has been the most active in studying usage statistics. Since February 2025, they have released five major Economic Index reports through its Economic Futures Programme.[37] OpenAI's Public Affairs team launched a public data initiative called OpenAI Signals in February 2026, which provides de-identified datasets covering adoption trends, regional diffusion, and work-related usage.[38] As of April 2026, other major labs like Google Gemini, xAI, and Meta, have not published similar usage statistics.

5.1.2 Why TechStat Needs Multi-lab Coverage

By establishing data-sharing agreements with AI labs, TechStat can gain near-real-time visibility into promising early signals of where displacement pressures are building. In practical terms, Anthropic is the most viable near-term data partner for TechStat. Over time, TechStat should seek broader coverage from additional labs as they develop their own reporting capabilities, because each platform's user base has distinct ways in which they use AI. For example, roughly one-third of Claude's work-related interactions involve coding, compared with approximately 4% of ChatGPT messages.[39] Chapter 6 discusses the international efforts currently underway to standardize AI usage reporting across labs, and Mila's role in ensuring that TechStat is positioned to receive and act on these statistics as they become available.

5.1.3 The Open-Weight Coverage Gap

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[36] Appel et al., "Anthropic Economic Index Report: Economic Primitives."
[37] Anthropic, "Economic Futures Program," Anthropic, accessed April 7, 2026, https://www.anthropic.com/economic-futures.
[38] OpenAI, Measuring AI Adoption, Protecting Privacy, and Empowering Decisions: OpenAI Signals Global Report (San Francisco: OpenAI, February 2026), https://cdn.openai.com/signals/openai-signals-global-report.pdf.
[39] Aaron Chatterji, Thomas Cunningham, David J. Deming, Zoë Hitzig, Christopher Ong, Carl Yan Shan, and Kevin Wadman, "How People Use ChatGPT," NBER Working Paper No. 34255 (September 2025), https://cdn.openai.com/pdf/a253471f-8260-40c6-a2cc-aa93fe9f142e/economic-research-chatgpt-usage-paper.pdf.

AI usage statistics from frontier AI labs carry structural coverage limitations that TechStat must account for. When a company or developer downloads an AI model and runs it on their own hardware, the lab that built the model has no visibility into how it is being used. These publicly available models (referred to as "open-weight" models, and including Meta's Llama, Mistral, Alibaba's Qwen, and DeepSeek, among others) generate no usage data that could flow into TechStat's monitoring system through a lab data-sharing agreement. Two independent studies published in late 2025 provide a sense of the size of this blind spot.

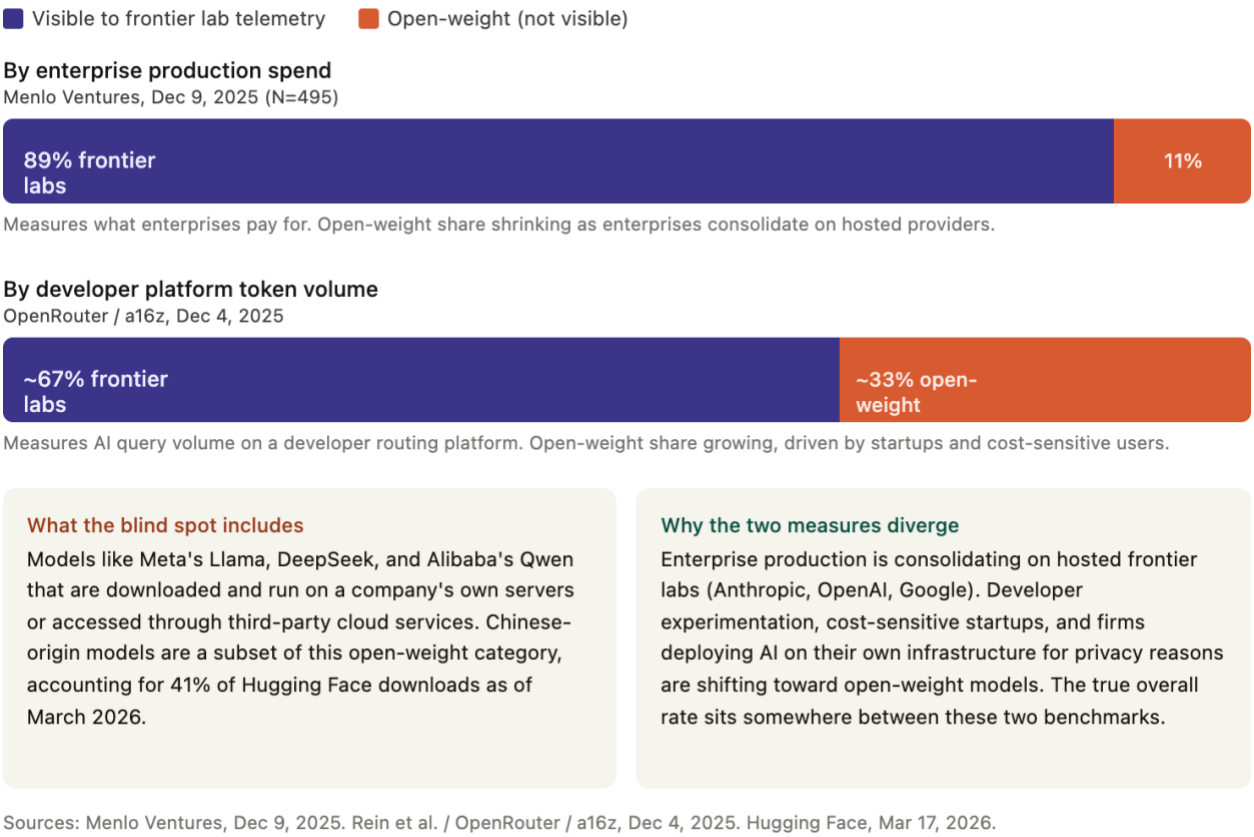

Figure 6. Coverage gap: frontier lab visibility by measurement method

Among large enterprise buyers, spending is consolidating with three frontier labs (Anthropic, OpenAI, and Google), and the open-weight share of enterprise production budgets has actually declined, from 19% in late 2024 to 11% by late 2025.[40] Among developers, startups, and

[40] Menlo Ventures, "2025: The State of Generative AI in the Enterprise," December 2025, https://menlovc.com/perspective/2025-the-state-of-generative-ai-in-the-enterprise/. The Menlo survey (approximately 500 U.S. enterprise decision-makers) places open-source models at 11% of enterprise LLM spend, down from 19% the prior year.

organizations that run AI on their own infrastructure for cost or privacy reasons, the trajectory runs in the opposite direction. Open-weight models reached roughly one-third of total processing volume on the OpenRouter platform by late 2025, up from single digits a year earlier.[41] Chinese-origin models alone account for 41% of downloads on Hugging Face, the world's largest public repository for AI models, surpassing U.S.-origin models for the first time as of spring 2026.[42]

5.1.4 Design Implication: Build on Domestic Data First

The practical implication for TechStat is that a monitoring system built solely on frontier lab data would miss an estimated 20 to 30% of total AI usage. Even within the 70 to 80% that frontier labs can see, acquiring the data requires voluntary cooperation from companies that have no obligation to share it. As discussed in Chapter 6, the international standardization effort currently underway is unlikely to produce a formal data-sharing commitment with even a single lab before early 2027. Given these two constraints, TechStat should build the monitoring system on the three domestic data streams first, create integration points for AI usage statistics telemetry, and treat lab data as an upgrade to a system that already functions independently. Pre-built monitoring infrastructure also lowers the barrier for labs to participate. Sympathetic staff inside frontier companies face lower internal political costs when they can advocate for connecting to a system that already exists rather than proposing that something new be built from scratch.[43]

5.2 Dynamic Task Exposure Metrics

Exposure metrics provide a theoretical assessment of how susceptible a given occupation is to AI automation. As established in Chapter 3, theoretical exposure scores cast too wide a net on their own and their value is to complement and contextualize AI usage statistics. For example, if AI usage statistics show a rise in automation-oriented usage for coding-related tasks, exposure

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[41] David Rein et al., "State of AI: An Empirical 100 Trillion Token Study with OpenRouter," arXiv:2601.10088, January 2026, https://arxiv.org/html/2601.10088v1. The study reports open-weight models at roughly one-third of processing volume on the OpenRouter platform by late 2025, up from single digits a year earlier.
[42] Hugging Face, "State of Open Source on Hugging Face: Spring 2026," Hugging Face Blog, 2026, https://huggingface.co/blog/huggingface/state-of-os-hf-spring-2026.
[43] The design principle of building monitoring infrastructure independently of lab cooperation draws on a conversation with Professor Jonathan Zittrain (Harvard Law School, Berkman Klein Center for Internet and Society), April 2026.

metrics help identify which occupations are most affected based on how prominent coding is within that occupation.[44]

TechStat should be aware that Statistics Canada's existing AI Occupational Exposure Index (C-AIOE) is anchored in a pre-generative AI methodology from 2021 and does not capture the capabilities of modern large language models.[45] Because model capabilities are advancing rapidly, a static exposure metric becomes outdated quickly. More recent research by Eloundou et al. demonstrated that frontier AI models can reliably score task-level exposure, and produce results that closely match with human expert panels.[46] This means TechStat can periodically re-score the full National Occupational Classification system using the latest available frontier models and track how exposure shifts over time. This requires no data-sharing agreements with AI labs and can be done internally.

Consultations conducted for this report suggest that Canada's statistical establishment has a strong institutional attachment for native measurement instruments (e.g., the Labour Force Survey and the C-AIOE exposure measure). Introducing a new exposure metric may face institutional resistance. Mila should anticipate this and be prepared to make the case for dynamic exposure scoring.

5.3 High-Frequency Administrative Payroll Data

AI telemetry and exposure scores together identify where automation pressure is building, but payroll data is needed to confirm that displacement has reached actual workers. High-frequency administrative payroll data refers to anonymized, transaction-level records generated by payroll companies like ADP and Dayforce through the processing of employee compensation. The 'Canaries in the Coal Mine' paper relied on exactly this kind of data to validate the link between automation-oriented AI usage and employment declines discussed in Chapter 3.[47]

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[44] Tyna Eloundou, Sam Manning, Pamela Mishkin, and Daniel Rock, "GPTs Are GPTs: An Early Look at the Labor Market Impact Potential of Large Language Models," arXiv preprint arXiv:2303.10130, March 2023, https://arxiv.org/abs/2303.10130.
[45] Tahsin Mehdi and René Morissette, "Experimental Estimates of Potential Artificial Intelligence Occupational Exposure in Canada," Analytical Studies Branch Research Paper Series, Statistics Canada Catalogue no. 11F0019M (September 2024), https://www150.statcan.gc.ca/n1/pub/11f0019m/11f0019m2024005-eng.htm.
[46] Eloundou et al., "GPTs Are GPTs."
[47] Brynjolfsson, Chandar, and Chen, "Canaries in the Coal Mine?"

5.3.1 What Payroll Data Captures That Surveys Cannot

Payroll data captures rates of hiring (job starts) and firing (separations) in near real-time. When new payroll accounts stop being created within a given employer or occupational category, that indicates a hiring freeze. When an existing payroll account stops receiving deposits, that indicates a termination. In contrast, Canada's LFS measures the stock of employed workers at a point in time and cannot distinguish between stable employment driven by high turnover and stable employment driven by zero hiring and zero firing.[48]

The difference in scale and granularity is another significant difference. As previously noted in Chapter 3, LFS surveys approximately 65,000 households per month, which translates to roughly 100,000 individuals.[49] Once that sample is sliced by occupation, age, province, and seniority, cell sizes become too small to be statistically significant. ADP Canada and Dayforce alone process payroll for roughly 90,000 Canadian client organizations, covering an estimated 40% of the workforce. That figure rises further when other commercial providers are included.[50]

The steps to acquiring this data and securing data partnerships with commercial providers are discussed in Chapter 6.

5.3.2 Coverage and Known Biases

Commercial payroll data covers an estimated 80% of the Canadian workforce and excludes self-employed individuals, independent contractors, and gig workers who do not file T4 slips, along with micro-businesses that tend to not use enterprise payroll platforms.[51] The monitoring needs of

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[48] Statistics Canada, Guide to the Labour Force Survey, 2025, Catalogue no. 71-543-G (Ottawa: Statistics Canada, 2025), https://www150.statcan.gc.ca/n1/pub/71-543-g/71-543-g2025001-eng.htm.
[49] Statistics Canada, "Labour Force Survey."
[50] ADP, "Payroll in Canada," Payroll Around the World (podcast), August 29, 2025, https://rethinkq.adp.com/payroll-podcast/payroll-in-canada/; ADP Canada, "Payroll and HR ROI," Articles and Insights, accessed April 12, 2026, https://www.adp.ca/en/resources/articles-and-insights/articles/p/payroll-and-hr-roi.aspx; Dayforce, "Ceridian to Accelerate Hiring in Canada with an Additional 2,000 Research, Development and Engineering Jobs" (newsroom press release, February 10, 2020), https://www.dayforce.com/who-we-are/newsroom/ceridian-to-accelerate-hiring-in-canada-with-an-additional-2,000-research-development-and-engineer.
[51] Staffing Industry Analysts, "Canada's Self-Employed at 13.2% of Working Population; Gig Percentage Slips," Global Daily News, June 4, 2024, https://www.staffingindustry.com/news/global-daily-news/canadas-self-employed-132-working-population-gig-percentage-slips; Statistics Canada, "Table 14-10-0215-01: Employment for All Employees by Enterprise Size, Annual," accessed April 12, 2026, https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=1410021501.

these excluded populations, and the extent to which the LFS can serve as a suitable tool for tracking them are discussed in **Appendix C**.

Within the 80% that payroll data does cover, the client base for any one payroll provider is not perfectly representative of the Canadian economy. The Canaries in the Coal Mine paper acknowledged that ADP's U.S. client base somewhat overrepresents manufacturing and services firms, certain geographic regions, and firms that tend to grow faster than average.[52] No study of ADP Canada's or Dayforce's client composition has been published, so the specific biases in the Canadian context are not yet documented. Generally, ADP and Dayforce serve mid-to-large employers, while smaller businesses tend to use Canadian platforms like Wagepoint and Knit People.

If future disruption patterns extend to sectors or firm sizes underrepresented in commercial payroll data, the early warning system's detection capability would be weaker in those areas. An effective monitoring system would ideally draw from both enterprise and SMB-level providers to approximate national representativeness.

5.4 Firm-Level AI Adoption Surveys

Firm-level AI adoption surveys can be a useful complement to the three data streams described above. Even when AI usage statistics, exposure metrics, and payroll data all point in the same direction, attributing an observed employment change to AI rather than to other economic forces is a persistent challenge. Interest rate shifts, immigration-driven supply changes, and sector-specific corrections can all produce similar-looking signals.

Statistics Canada demonstrated the feasibility of collecting this data through the Canadian Survey on Business Conditions (CSBC), which introduced AI adoption questions in Q2 2024. The survey asked firms whether they used AI to produce goods or deliver services, which types of AI they used, and whether adoption changed their employment levels or employee tasks.[53] However, the CSBC is a rapid-response survey designed to rotate topics based on emerging economic conditions, and its

[52] Brynjolfsson, Chandar, and Chen, "Canaries in the Coal Mine?"
[53] Valerie Bryan, Shivani Sood, and Chris Johnston, "Analysis on Artificial Intelligence Use by Businesses in Canada, Second Quarter of 2025," Analysis in Brief, Statistics Canada Catalogue no. 11-621-M, no. 2025008 (Ottawa: Statistics Canada, 2025), https://www150.statcan.gc.ca/n1/pub/11-621-m/11-621-m2025008-eng.htm.

AI module has not been fielded every quarter. Statistics Canada also announced that the CSBC's final collection cycle will take place in summer 2026 and the final release is scheduled for August 2026.[54] This means the government is about to lose its only high-frequency source of firm-level AI adoption data at the exact moment when TechStat is being set up.

TechStat will need to secure an alternative quarterly vehicle for firm-level AI adoption data, and Chapter 6 will discuss specific actions Mila can take to help TechStat do so.

5.5 Preparing Federal and Provincial Stakeholders to Act on This Data

Mila should use the period before the data arrives to build analytical readiness across the institutions that will need to use the data or act on the data.

ACTION STEPS

- [M] **Mila** should brief TechStat directly on the core concepts of this report including the automation-augmentation distinction, the limits of static exposure metrics relative to dynamic task exposure metrics, and the value of high-frequency payroll data.
- [M] **Mila** should engage provincial labour market officials to ensure they understand what the early warning system will produce and what kinds of decisions they can inform.

[54] Statistics Canada, "The Daily: Canadian Survey on Business Conditions, First Quarter 2026," February 27, 2026, https://www150.statcan.gc.ca/n1/daily-quotidien/260227/dq260227c-eng.htm.

CHAPTER 06

Acquiring the Data

CHAPTER AT A GLANCE

This chapter addresses how TechStat acquires each data stream, including which institutions to engage, which legal instruments to use, and what timelines to expect.

Domestic survey instruments are already operational within Statistics Canada but require specific actions to preserve and adapt them for the early warning system. The Labour Force Survey is the most representative labour market instrument in Canada and should serve as a broad corroborative signal while payroll partnerships are being established. TechStat should request that the LFS digital platform supplement move from annual to quarterly frequency. The more urgent challenge is the Canadian Survey on Business Conditions. Its final collection cycle is scheduled for August 2026, and if TechStat does not absorb the CSBC's AI module before it sunsets, Canada will lose its only high-frequency source of firm-level AI adoption data with no replacement in place.

Payroll data partnerships with ADP Canada and Dayforce should be pursued simultaneously. Each provider covers roughly 20% of the Canadian workforce, and ADP's existing research partnership with Stanford's Digital Economy Lab demonstrates a willingness to share anonymized payroll data. The voluntary partnership model is the correct approach. Statistics Canada's 2018 attempt to compel financial data under Section 13 triggered public backlash, a Privacy Commissioner investigation, and a temporary halt on relevant projects. Federal public sector payroll data carries a separate challenge: the Phoenix pay system still has over 233,000 unresolved transactions affecting 133,000 public servants, and the transition to Dayforce is not expected to be complete until March 2031. TechStat should treat federal payroll records as lower-confidence inputs when integrating them into the early warning system.

AI usage statistics from frontier labs are the highest-value but slowest-to-arrive data stream. The international standardization effort led by Sam Manning's team faces a coordination problem. Each lab classifies usage differently, and no company wants to adopt a competitor's framework as the standard. A realistic timeline for a formal data-sharing commitment with even a single lab (likely Anthropic) is early 2027. Canada should support the multilateral effort rather than pursue bilateral requests, and Mila's role is to ensure the reporting standard meets Canadian needs, including the

> *appropriate level of geographic and occupational disaggregation. If Anthropic is the first lab to share data, TechStat should be aware that Claude's user base skews heavily toward coding and technical work (37% of interactions) compared with ChatGPT (4.2%), and consider how to adjust for this. The SOC-to-NOC occupational crosswalk has not been updated since 2020 and will need to be operational before the first telemetry data arrives.*

6.1 Domestic Survey Instruments

6.1.1 Labour Force Survey (LFS)

While the limitations of LFS have been acknowledged, it is a useful data stream to integrate in the short-term while data partnerships with Payroll companies are being established. Tyler Meredith, former Lead Economic Advisor to Prime Minister Justin Trudeau, identified the LFS as the starting point for any AI monitoring effort, noting that no other instrument has comparable representativeness across the Canadian economy. Additionally, increasing the digital platform supplement's frequency from annual to quarterly would improve TechStat's ability to track directional trends among gig workers. Long-term, the LFS is more likely to serve as a broad corroborative signal than as a precise detection instrument.

ACTION STEPS — §6.1.1

[T] **TechStat** should use the LFS and request that the digital platform supplement be moved from an annual to a quarterly frequency. This is a low-cost action that can demonstrate the functionality of the system while payroll data becomes available. Changes to LFS survey instruments require approval from the Chief Statistician, André Loranger.[55]

6.1.2 Securing a Replacement to the CSBC

As discussed in 5.4, Statistics Canada has announced that the CSBC's final collection cycle will take place in August 2026.[56] There are three options for preserving this data stream:

1. **Absorb the CSBC AI Module:** TechStat advocates to take over the existing module under its own mandate. This is the most feasible option because the methodology and

[55] Government of Canada, "André Loranger, Chief Statistician of Canada (Interim)," accessed April 12, 2026, https://www.canada.ca/en/government/management/andre-loranger.html.
[56] Statistics Canada, "The Daily: Canadian Survey on Business Conditions, Q1 2026."

infrastructure already exist. If TechStat acts before the survey sunsets, it could continue fielding the same questions with minimal interruption.

2. **Migrate to the Survey of Innovation and Business Strategy (SIBS):** The Survey of Innovation and Business Strategy (SIBS) is the most natural candidate since it already covers AI adoption practices. Meredith identified it as the instrument most likely to produce granular data. However, SIBS runs on a multi-year cycle (most recently in 2022), which would need to change.[57] Changing the survey's multi-year cycle is significantly less feasible and would likely take beyond August 2026.

3. **Develop a Purpose-Built Survey:** A purpose-built survey would be the most analytical power option but would likely need 12 to 18 months to design, pilot, and field.[58]

ACTION STEPS — §6.1.2

[T] TechStat should advocate to absorb the CSBC AI module before August 2026, and pursue either of the other options if this option fails or is insufficient.

[M] Mila should flag CSBC discontinuation to TechStat as a near-term data risk, and support TechStat through Innovation, Science and Economic Development Canada (ISED) if needed.

- Mila can reinforce the case for AI-adoption surveys, if needed, by raising the issue with Innovation, Science and Economic Development Canada (ISED), which Statistics Canada reports into. Tyler Meredith emphasized that a signal from the Minister of Innovation to the Chief Statistician carries significant institutional weight. The request aligns with Statistics Canada's own stated direction: StatsCan's 2026-27 Departmental Plan commits the agency to emphasizing technology-focused surveys to capture the impact of AI on Canadian businesses and households.[59] Framing the request around the government's own stated commitments increases StatsCan's justification to act.

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[57] Statistics Canada, "Survey of Innovation and Business Strategy (SIBS)," Surveys and Statistical Programs, Record no. 5171, last modified February 20, 2024, https://www23.statcan.gc.ca/imdb/p2SV.pl?Function=getSurvey&SDDS=5171.

[58] Andrew Heisz, Evaluation of the Canadian Household Panel Survey Pilot Content, Statistics Canada Catalogue no. 89-648-X (Ottawa: Statistics Canada, 2013), https://www150.statcan.gc.ca/n1/pub/89-648-x/89-648-x2013001-eng.htm.

[59] Statistics Canada, Departmental Plan 2026-2027 (Ottawa: Statistics Canada, 2026), https://www.statcan.gc.ca/en/about/transparency/departmental-plan/departmental-plan-2026-2027.pdf.

> *"If the Minister of Industry says, 'We need faster and better answers on this, and we need you to engage with Mila,' that as a nudging function in government will be very important."*
>
> — Tyler Meredith, former Lead Economic Advisor to Prime Minister Justin Trudeau

6.2 Payroll Data Partnerships

TechStat should establish voluntary data-sharing agreements with commercial payroll providers like ADP Canada, Dayforce, and smaller providers like Wagepoint that serve small and medium-sized businesses.

Which providers to approach: TechStat should prioritize ADP Canada and Dayforce, and approach them simultaneously. ADP Canada serves one in five Canadian workers and ADP's research partnership with Stanford's Digital Economy Lab demonstrates a willingness to share granular anonymized payroll data for academic research.[60] Dayforce also serves roughly 20% of Canadian workers, and has an existing Government of Canada relationship and is replacing the Phoenix pay system.[61] Dayforce has no precedent for external data-sharing but its existing engagement with the federal government creates a foundation for engagement.

The case for a voluntary partnership: Statistics Canada has the legal authority to compel private-sector data under Section 13 of the Statistics Act, which allows the Chief Statistician to require any organization to provide information for statistical purposes.[62] However, precedent strongly argues against using this approach. In 2018, Statistics Canada used Section 13 to compel banks and credit bureaus to provide detailed individual financial transaction data. The initiative triggered substantial public backlash, intense media scrutiny, parliamentary hearings, and a formal investigation by the Office of the Privacy Commissioner. The Commissioner found that while Statistics Canada had the legal authority, the approach raised concerns about necessity and

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[60] ADP, "Payroll in Canada;" Brynjolfsson, Chandar, and Chen, "Canaries in the Coal Mine?"
[61] Public Services and Procurement Canada, "Government of Canada Moves Forward with HR and Pay Transformation through Dayforce" (news release, June 2025), https://www.canada.ca/en/public-services-procurement/news/2025/06/government-of-canada-moves-forward-with-hr-and-pay-transformation-through-dayforce.html.
[62] Statistics Canada, "Generic Privacy Impact Assessment," Principle 3: Consent, accessed April 7, 2026, https://www.statcan.gc.ca/en/about/transparency/privacy-impact-assessments/generic/section5.

proportionality under the Privacy Act.[63] A detailed walkthrough of the relevant legal provisions (PIPEDA Section 7(3)(f), Statistics Act Section 17, the Administrative Data Acquisition Agreement vehicle) is provided in **Appendix D**.

ACTION STEPS — §6.2

[M] Brief TechStat on payroll partnership technical and legal considerations. (§6.2)

[T] Secure voluntary data-sharing partnerships with ADP Canada and Dayforce with the goal of beginning monthly data deliveries by early 2027.

6.2.1 Public Sector Payroll Data Challenges

A significant portion of Canadian government employment has historically relied on a fragmented mix of proprietary and legacy systems and not covered by major payroll providers. The Federal Government of Canada uses the internal (and infamous) Phoenix Pay System to process pay for over 430,000 public servants which sits outside commercial payroll aggregators and currently cannot serve as a reliable analytical source due to significant data integrity issues. As of late 2025, over 233,000 transactions affecting more than 133,000 public servants remained unresolved in Phoenix.[64] The categories of transactions most affected by Phoenix errors, including terminations, status changes, and leave returns, are precisely the categories relevant for the early warning system.[65]

The federal government is currently transitioning from Phoenix to Dayforce, with full implementation expected by March 2031. In March 2026, Auditor General Karen Hogan warned that

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[63] CBC News, "Privacy Commissioner Launches Probe into StatsCan Over Collection of Financial Data," CBC News, October 26, 2018, https://www.cbc.ca/news/politics/personal-financial-information-statistics-canada-1.4885945; Office of the Privacy Commissioner of Canada, "Investigation into Statistics Canada's Collection of Financial Transactions Information" (PA-20191209-SC, December 9, 2019), https://www.priv.gc.ca/en/opc-actions-and-decisions/investigations/investigations-into-federal-institutions/2018-19/pa_20191209_sc/.

[64] Public Services and Procurement Canada, "The Government of Canada Provides Update on Progress toward a New Human Resources and Pay Solution to Replace the Phoenix Pay System and Its Human Resources Systems" (news release, July 2024), https://www.canada.ca/en/public-services-procurement/news/2024/07/the-government-of-canada-provides-update-on-progress-toward-a-new-human-resources-and-pay-solution-to-replace-the-phoenix-pay-system-and-its-human-.html; David Fraser, "Auditor General Flags Potential Risks in Transition from Phoenix to Dayforce," CBC News, March 23, 2026, https://www.cbc.ca/news/canada/ottawa/phoenix-dayforce-federal-canada-government-pay-9.7138444.

[65] Public Service Alliance of Canada, "Government Downplays Phoenix Pay System Failures while Workers Continue to Suffer Nine Years Later" (news release, February 27, 2025), https://psacunion.ca/government-downplays-phoenix-pay-system-failures.

if these issues are not cleared before the transition, the existing data integrity issues could also affect the new system.[66]

Therefore, TechStat should treat federal payroll records as lower-confidence inputs and seek ways to cross-reference any signals against independent sources. The likely correct approach is to still utilize the available data, but with transparent disclosures that make clear what share of the federal workforce meets data quality thresholds, rather than ignoring the data completely or overstating its reliability. A detailed protocol for how TechStat should manage this issue is beyond the scope of this report. Instead, this is flagged as a known implementation challenge that TechStat will need to address as it builds out the monitoring framework.

ACTION STEPS — §6.2.1

[M] Advise TechStat on Phoenix data limitations as a source of public sector payroll data. (§6.2.1)

6.2.2 Potential Alternative: ePayroll Initiative

If TechStat is looking for a government-native alternative to commercial payroll partnerships, the closest existing initiative is the federal ePayroll project.[67] The purpose of this project is to modernize CRA's tax and benefits administration, and it would require employers to submit individual-level payroll data to CRA at the time of each pay run in place of annual T4 filings. Phase 1 of ePayroll was completed by March 2024, but Phase 2 (implementation) has not yet received dedicated federal funding and a full national rollout is not expected in the near term.[68]

The ePayroll project would provide individual-level identification of workers at high frequency, and allow TechStat to detect new hires and separations in near-real-time. However, as currently designed, there is no indication in the available Phase 1 documentation that mandatory standardized occupational codes are a planned field, which would leave TechStat blind to the

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[66] Fraser, "Auditor General Flags Potential Risks."
[67] Canada Revenue Agency, "ePayroll Project," Government of Canada, accessed April 12, 2026, https://www.canada.ca/en/revenue-agency/corporate/about-canada-revenue-agency-cra/epayroll-project.html.
[68] Canada Revenue Agency, 2024-25 Departmental Plan (Ottawa: CRA, 2024), https://www.canada.ca/en/revenue-agency/corporate/about-canada-revenue-agency-cra/departmental-plan/2024-25-cra-departmental-plan.html; National Payroll Institute, Canada's Need for ePayroll (Toronto: National Payroll Institute, February 2025), https://payroll.ca/CPAWebsite/media/PDF/Priority%20Issues/February-2025-Canada-s-need-for-ePayroll-National-Payroll-Institute.pdf.

occupation of individuals. If TechStat decides to pursue this pathway, it should ensure that the National Occupational Classification is integrated into the system.

6.3 AI Usage Statistics from Frontier Labs

According to Sam Manning, the biggest open design problem is how labs can share firm-level usage patterns at a useful aggregation level (by industry, firm size, and geography) without identifying the firms themselves.

Manning described a realistic timeline of approximately one year (early 2027) before a standardized reporting commitment with even a single AI lab could plausibly land (likely Anthropic). Besides having to identify the right reporting template, Manning and his team face a coordination problem: each lab classifies AI usage differently and no company wants to adopt a competitor's framework as the default. Ketana Krishna, an independent contractor with the Oxford Martin AI Governance Initiative involved with Manning's efforts, emphasized that frontier labs are generally open to sharing usage statistics, but conversations consistently stall at the standardization stage.

In consultation with Mila, this report concluded that Canada and Mila should support this multilateral effort instead of pursuing bilateral data requests independently. Labs are more likely to commit to a single standardized reporting format serving multiple countries than to negotiate separate arrangements country by country. Mila's role in this process is to ensure the international reporting standard meets Canada's needs, and prepare the federal and provincial officials who will use it.

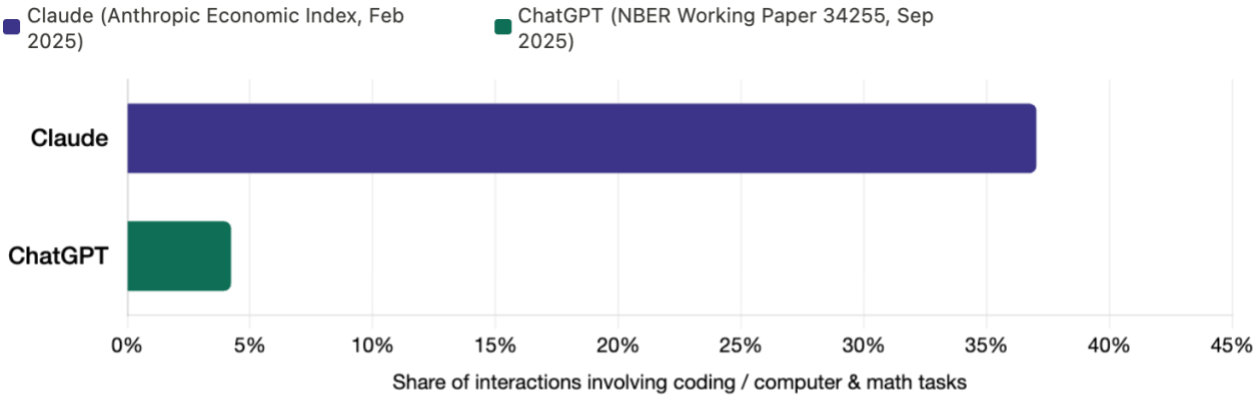
Ensuring the AI usage statistics meet Canada's needs: The level of disaggregation provided by Manning's reporting standard will inform how useful this data is. Provincial governments may also have input on the level of granularity they need to act on the signals, and Mila should consult with provincial labour market ministries to incorporate their requirements into the data specification.

The reporting standard Manning's team is designing will likely default to U.S. O*NET task descriptions and SOC occupation codes, which TechStat would map onto Canada's National Occupational Classification (NOC). The existing crosswalk between SOC and NOC has not been updated since 2020 and is based on NOC 2016 instead of the current NOC 2021 framework.[69]

[69] LMIC, "Job-Skills Mapping."

TechStat should ensure that a workable translation logic is available by the time AI usage statistics arrive.

The single-lab skew problem: If only one lab participates in data-sharing initially (most likely Anthropic, given its track record of publishing economic usage data), the resulting signal would carry a measurable skew. Anthropic's Economic Index reports that roughly 37% of Claude interactions involve computer and mathematical tasks.[70] By comparison, a September 2025 study of ChatGPT usage found that computer programming accounted for approximately 4.2% of ChatGPT messages, and the authors explicitly attributed the gap to "the difference in types of users."[71] TechStat should be aware of this skew and consider how to adjust for it.



Divergence factor
~9x

Implication
An Anthropic-only signal would overstate automation pressure in software and analytical roles, while understating shifts in customer service, healthcare, and education.

Note: The two measures are not identical. Anthropic reports the share of interactions classified under "computer and mathematical" occupational tasks. The ChatGPT figure captures messages classified as "computer programming." The NBER paper's authors explicitly contrast the two figures and attribute the divergence to differences in user base composition.

Figure 7. Coding task share by platform: single-lab occupational skew

[70] Anthropic, "Introducing the Anthropic Economic Index," February 10, 2025, https://www.anthropic.com/news/the-anthropic-economic-index.
[71] Aaron Chatterji et al., "How People Use ChatGPT," NBER Working Paper 34255, September 2025, https://www.nber.org/system/files/working_papers/w34255/w34255.pdf.

ACTION STEPS — §6.3

[M] Mila should brief TechStat on why the early warning system should be designed to operate on domestic data streams first, with frontier lab telemetry integrated as a plug-in upgrade when available, so that the system's functionality does not depend on international standardization timelines.

[T] TechStat should assess the current state of the SOC-to-NOC translation pathway, including interim workarounds (the NOC 2016-to-2021 concordance tables, the Dais's third-party crosswalk) and the timeline for ESDC's Occupational and Skills Information System (OaSIS), which is intended to serve as Canada's equivalent to O*NET and would reduce the need for international crosswalks in the future.[72] If OaSIS will not be operational in time for the first telemetry data deliveries, TechStat should identify which interim crosswalk solution it will use and confirm that it is adequate for the system's needs.

[M] Mila should consult with provincial labour market ministries through the Forum of Labour Market Ministers (FLMM) and bilaterally to understand their data needs, including the appropriate level of geographic disaggregation, and communicate Canada's data requirements with Manning's team. The implementation section for Recommendation 3 details the engagement plan, combining bilateral outreach to Ontario and Quebec with a pan-Canadian channel through FLMM's Labour Market Information Council (LMIC).

[72] Employment and Social Development Canada, "National Occupational Classification (NOC)," Government of Canada, accessed April 12, 2026, https://www.canada.ca/en/employment-social-development/services/noc.html; Viet Vu, "The O*NET/NOC Crosswalk: An Update," The Dais at Toronto Metropolitan University, November 2022, https://dais.ca/reports/crosswalk-blog-post/; Statistics Canada, "NOC 2016–NOC 2021 Concordance."

CHAPTER 07

Preparing the Response

CHAPTER AT A GLANCE

Canada has not yet designed policy responses tailored to AI-driven labour displacement. This chapter draws lessons from the COVID-19 emergency response, surveys international policy discussions that are moving fastest, and identifies what Canada needs to do now to avoid repeating the mistakes of 2020.

Three findings from the CERB experience are relevant. First, ESDC's benefits delivery infrastructure could not handle the crisis. Legacy systems running on COBOL mainframes could not process the surge in claims or adapt to new eligibility criteria, and the government bypassed ESDC entirely by routing CERB through CRA. Any AI-specific benefit requiring new eligibility parameters would likely face a 6-to-18-month implementation timeline under current infrastructure, which means benefit designs need to exist before the emergency that triggers them. Second, the government could not verify employment status in real time. CRA's most recent income data at the time of CERB's launch was from the 2018 tax year, and the decision to process applications on an attestation basis contributed to $4.6 billion in confirmed overpayments and at least $27.4 billion in payments flagged for further investigation. The Auditor General explicitly recommended developing a real-time payroll system. Third, CERB payments triggered automatic claw-backs against provincial social assistance in several provinces because the federal government had no time to negotiate exemptions with 13 provincial and territorial social services ministries before launch. Pre-crisis agreements on how a federal AI displacement benefit would be treated under provincial means-testing rules would prevent this from recurring.

The international policy community is moving quickly on AI displacement preparedness. The Forecasting Research Institute surveyed experts on six policy proposals across slow, moderate, and rapid AI scenarios from mid-October 2025 to the end of February 2026. Windfall Trust convened over 70 senior experts in London in February 2026 for a structured scenario exercise, then published a Policy Atlas cataloguing 48 proposals in April 2026. The Partnership on AI launched a Labour and Economy Steering Committee built on Windfall's work by the end of March 2026. None of these initiatives have been translated into Canada's specific institutional context: its Employment Insurance

> *architecture, its Labour Market Transfer Agreement structure, or the federal-provincial division of responsibility for workforce adjustment. Mila should lead a Canadian scenario exercise that maps plausible AI displacement scenarios to candidate policy responses and produces a working document that gives ESDC a starting point for designing interventions Canada can deploy.*

7.1 Lessons from Canada's COVID-19 Economic Response

On March 11, 2020, the World Health Organization declared COVID-19 a global pandemic. Between March 12 and March 22, every Canadian province and territory declared a state of emergency.[73] Mandated closures of all non-essential businesses followed soon after and triggered the sharpest labour market contraction in the country's history. Between February and April 2020, Canada lost 3 million jobs and experienced a 15.7% contraction that Statistics Canada described as far exceeding any previous downturn.[74]

The federal government's flagship response, the Canada Emergency Response Benefit (CERB), moved from announcement to issuing its first payments in under three weeks. Applications opened on April 6, 2020, and Canadians with direct deposit received payments within three business days.[75] This speed came at a price, and the trade-offs the government made under pressure reveal structural vulnerabilities that could resurface in an AI displacement scenario.

Three findings are particularly relevant for this report:

1. The government's primary benefits system could not handle the crisis: During COVID-19, the volume of unemployment claims overwhelmed ESDC's Employment Insurance (EI) infrastructure. The Office of the Auditor General of Canada (OAG) reported that more than 6 million CERB and EI applications were received between March 15 and April 19, 2020.[76] ESDC's benefits delivery systems, some over 50 years old, run on COBOL mainframes and could not process the surge or

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[73] Savannah Sewell, "COVID-19 in Canada: A Timeline," Museum of Health Care at Kingston (blog), n.d., accessed April 12, 2026, https://www.museumofhealthcare.ca/blog/covid-19-in-canada-a-timeline.
[74] Statistics Canada, "The Daily: Labour Force Survey, April 2020," May 8, 2020, https://www150.statcan.gc.ca/n1/daily-quotidien/200508/dq200508a-eng.htm.
[75] Employment and Social Development Canada, "Canada Emergency Response Benefit to Launch on April 6" (news release, April 2020), https://www.canada.ca/en/employment-social-development/news/2020/04/canada-emergency-response-benefit-to-launch-on-april-6.html.
[76] Office of the Auditor General of Canada, "Report 6: Canada Emergency Response Benefit," in 2021 Reports of the Auditor General of Canada to the Parliament of Canada (Ottawa: OAG, March 2021), https://www.canada.ca/en/auditor-general/our-work/audit-reports/parl-oag-202103-01-e.html.

implement changes to eligibility criteria in time.[77] The government bypassed ESDC entirely and routed CERB through the Canada Revenue Agency whose systems were optimized for mass, uniform disbursements.[78]

Even if an AI displacement event does not produce a similar sudden volume of requests, the key constraint is the rigidity of ESDC's legacy infrastructure. This system is difficult to reconfigure quickly for any new benefit design. For example, the 2021 EI Sickness Benefit extension was announced in the April 2021 Budget but was not implemented until December 18, 2022.[79] Any AI-specific intervention requiring new eligibility parameters or benefit designs would likely face a 6-to-18-month implementation timeline.

Key takeaway: If a novel AI displacement benefit needs to be designed during a mass labour displacement emergency, Canada's benefits delivery infrastructure is not set up to adapt in time and the government would likely deliver through CRA. The Canadian government should proactively design appropriate policy responses while there is lead time instead of waiting for an emergency that creates a rushed response. The early warning system itself is a tool for giving the government advance notice and extending the window for deliberate policy design.

2. The government could not verify eligibility in real-time: When CERB launched in April 2020, CRA's most recent income data came from the 2018 tax year as the 2019 filing season was still underway. CRA had no way to confirm whether applicants had experienced an income loss in March 2020 because no system existed to provide current employment or payroll information. The government decided to process applications on an attestation basis and defer eligibility confirmation to post-payment reconciliation.[80]

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[77] Employment and Social Development Canada, "HUMA Committee Appearance - Beech Briefing Binder, May 6, 2024," Government of Canada, https://www.canada.ca/en/employment-social-development/corporate/reports/committe-binders/2024-may-6-huma-beech.html; Office of the Auditor General of Canada, "Report 8—The Benefits Delivery Modernization Programme," in 2023 Reports of the Auditor General of Canada to the Parliament of Canada (Ottawa: OAG, 2023), https://www.canada.ca/en/auditor-general/our-work/audit-reports/parl-oag-202310-08-e.html.
[78] OAG Canada, "Report 6: Canada Emergency Response Benefit."
[79] Employment and Social Development Canada, "Question Period Note EWDDI June 2023 018" (Open Government, Government of Canada, June 2023), https://search.open.canada.ca/qpnotes/record/esdc-edsc%2CEWDDI_June2023_018.
[80] Canada Revenue Agency, "Standing Committee on Public Accounts - Briefing Binder, April 15, 2021," Government of Canada, https://www.canada.ca/en/revenue-agency/corporate/about-canada-revenue-agency-cra/transparency-proactive-disclosure-canada-revenue-agency/disclosure-briefing-binders-parliamentary-appearances/standing-committee-public-accounts-2021-04-15.html.

Based on OAG's 2022 audit, the federal government spent $210.7 billion on the COVID-19 benefits covered by the audit, including $74.8 billion on CERB and the EI Emergency Response Benefit. The OAG found $4.6 billion in confirmed overpayments to ineligible recipients and flagged at least $27.4 billion in payments requiring further investigation. As of summer 2022, only about half ($2.3 billion) was recovered. The 2022 audit stated it is likely that a significant amount of money will not be recovered. The OAG explicitly recommended that CRA and ESDC pursue development of a real-time payroll system to improve the efficiency of future benefit delivery.[81]

Key takeaway: The inability to verify employment status in real-time can carry a significant financial cost when responding to a labour emergency. The voluntary payroll data partnerships proposed in Chapter 6 are designed primarily as a detection tool and would likely not have the individual-level granularity or population-wide coverage needed for benefit eligibility verification. However, a mandatory reporting infrastructure such as the federal ePayroll initiative discussed in Chapter 6 would cover all employers and provide individual-level data. While the scale of the verification failures during COVID and the OAG's recommendation to pursue a real-time payroll system strengthen the case for the ePayroll initiative, privacy considerations and the potential for political backlash must be thoroughly considered.[82]

3. Federal benefits collided with provincial social assistance systems: Provincial social assistance programs in Canada are means-tested, and their systems automatically reduce or eliminate benefits when a recipient's income from other sources exceeds certain thresholds.[83] When CERB payments started arriving, these provincial systems treated CERB as income that reduced recipients' eligibility for provincial support and applied their pre-existing rules. Some provinces like Nova Scotia, Saskatchewan, and New Brunswick, applied dollar-for-dollar claw-backs against provincial social assistance and disability payments.[84] The net effect was that some of the most vulnerable Canadians saw no financial benefit from the federal intervention.

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[81] Office of the Auditor General of Canada, "Report 10: Specific COVID-19 Benefits."
[82] CRA, "ePayroll Project."
[83] Maytree, "Eligibility for Social Assistance: Assets and Income," Welfare in Canada, July 2025, https://maytree.com/changing-systems/data-measuring/welfare-in-canada/eligibility-for-social-assistance-assets-and-income/.
[84] Gillian Petit and Lindsay M. Tedds, "The Effect of Differences in Treatment of the Canada Emergency Response Benefit across Provincial and Territorial Income Assistance Programs," Canadian Public Policy 46, no. S1 (July 1, 2020): S29–S43, https://doi.org/10.3138/cpp.2020-054.

Because the federal government designed and launched CERB in three weeks, there was no time in that window to negotiate exemptions with 13 provincial and territorial social services ministries.[85] Some provinces eventually created full or partial exemptions, but the response was uneven and led to a patchwork of outcomes across the country.[86]

Key takeaway: If the federal government and the provinces agree before a crisis that a certain AI displacement benefit will be exempt from provincial means-testing, or that provinces will hold benefit levels constant for a defined period after a federal intervention begins, the claw-back problem will not repeat.

ACTION STEPS — §7.1

[M] Mila should brief TechStat on the COVID-19 case study findings in this report, focusing on the fiscal risks of deploying emergency benefits without real-time employment verification and the importance of pre-designed benefit parameters.

[E] ESDC should proactively identify suitable AI displacement benefit designs and their implementation requirements (eligibility rules, delivery channel, funding levels). The scenario-to-policy mapping exercise proposed in section 7.2 would produce a working document that informs this process.

[E] ESDC should establish proactive agreements with provinces on how a future federal AI displacement benefit would be treated under provincial means-testing rules.

7.2 Identifying the Right Policy Responses to AI Labour Shocks

Canada has not yet established policy responses for various AI displacement scenarios. Identifying the right policy responses for Canada requires a structured process that accounts for Canada's specific institutional constraints and existing policy toolkit. This section surveys the current state of expert consensus on AI displacement policy solutions and outlines how Mila can guide the creation of Canada's policy response toolkit.

[85] Cann v. Director, Fort Garry/River Heights, 2020 MBCA 101, at paras. 8–10 (Man. C.A.) (Mainella, J.A.), https://www.canlii.org/en/mb/mbca/doc/2020/2020mbca101/2020mbca101.html.
[86] Petit and Tedds, "Effect of Differences in Treatment of the CERB."

The current state of AI displacement policy discussions

Across several major research initiatives over the past year, economists, AI researchers, and policymakers have come together to analyze plausible future scenarios of AI-driven job displacement at different levels of speed and severity, and to assess which policy instruments are appropriate for each. The following examples illustrate the shape of these discussions:

- **The Forecasting Research Institute (FRI)** surveyed economists, AI experts, superforecasters, and the public on six policy proposals between October 2025 and February 2026. Proposals were evaluated against slow, moderate, and rapid AI progress scenarios. Retraining support programs received the widest endorsement (71.8% support), universal basic income was more divisive (37.4% support), while Compute taxes and job guarantee programs were the most opposed.[87]
- **Windfall Trust** convened over 70 senior economists, AI researchers, civil servants, and policy experts in London in **February 2026** for a structured scenario exercise. Participants were placed in a 2030 scenario where nearly a third of top earners have lost their jobs due to AI automation. Working in groups, participants provided options for the Prime Minister.[88]
    - Following the Windfall Trust's London exercise, the Partnership on AI launched a Labour and Economy Steering Committee on **March 30, 2026**, which will develop its recommendations by leveraging Windfall's scenario analysis.[89]
    - Windfall Trust published their Policy Atlas on **April 8, 2026**, a catalog of more than 48 policy proposals for responding to the economic disruption from transformative AI. The Atlas organizes proposals across five domains: Public and Social Investments, Labor Market Adaptation, Wealth Capture, Regulation and Market Design, and Global Coordination. Each policy is tagged by the rate of disruption it is

[87] Forecasting Research Institute, "Forecasting the Economic Effects of AI," FRI Substack, March 2026, https://forecastingresearch.substack.com/p/forecasting-the-economic-effects-of-ai.
[88] Danny Buerkli and Brandon Jackson, "When the Economy Meets AI: What Happens Next?" Windfall Trust Scenarios, n.d., accessed April 12, 2026, https://windfalltrust.org/scenarios/when-the-economy-meets-ai-what-happens-next.
[89] Partnership on AI, "Partnership on AI Launches Expert Advisory Group for New Initiative Shaping Economic Futures in the AI Era" (news release, March 30, 2026), https://partnershiponai.org/partnership-on-ai-launches-expert-advisory-group-for-new-initiative-shaping-economic-futures-in-the-ai-era/.

designed to address, distinguishing between a gradual scenario, a rapid scenario, and policies that apply across all scenarios.[90]

  - This Windfall Trust workshop built on insights from a prior September 2025 workshop led by Erik Brynjolfsson, Ajay Agrawal, and Anton Korinek. That earlier convening, hosted at Stanford, brought together a dozen research teams to present draft chapters for a forthcoming NBER volume on the economics of transformative AI, including the value of transformative AI, labour and wage dynamics under AGI, public finance, and the measurement challenges that AI creates for GDP measurement.[91]

- **Anthropic** published a policy exploration in October 2025 based on consultations with members of their Economic Advisory Council and Economic Futures Symposia. They reviewed proposals including workforce development, permitting reform and fiscal policy.[92]

It's important to note how fast this field is moving. Windfall Trust ran its London exercise in February 2026, the Partnership on AI launched a Labour and Economy Steering Committee built on that exercise by the end of March 2026, and the Windfall Policy Atlas was published in April 2026.

However, there is still a gap (and opportunity) to translate these proposals into specific national institutional contexts. As the FRI research shows, there are varying perspectives on the right policy responses, and while the Windfall Atlas catalogues 48 proposals across five domains, it does not assess which of those proposals are operable within Canada's Employment Insurance architecture, its Labour Market Transfer Agreement structure, or the federal-provincial division of responsibility.

Identifying Canada's response toolkit requires actors with the scientific and economic knowledge to assess which AI displacement scenarios are plausible, and the policy fluency to evaluate which

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[90] Windfall Trust, Windfall Policy Atlas, 2026, https://windfalltrust.org/policy-atlas.
[91] NBER, "Economics of Transformative AI Workshop," organized by Ajay K. Agrawal, Anton Korinek, and Erik Brynjolfsson, Stanford Institute for Economic Policy Research, September 18–19, 2025, https://www.nber.org/conferences/economics-transformative-ai-workshop-fall-2025; Tom Cunningham, "Economics and Transformative AI," personal blog, October 2, 2025, https://tecunningham.github.io/posts/2025-09-19-transformative-AI-notes.html.
[92] Anthropic, "Preparing for AI's Economic Impact: Exploring Policy Responses," Anthropic (research), October 14, 2025, https://www.anthropic.com/research/economic-policy-responses.

proposals fit Canada's institutional realities. Mila is very well positioned to lead an initiative like this. Conducting a Canadian scenario exercise translates the international policy work surveyed in this section into something Canada can deploy when TechStat begins surfacing early warning signals.

ACTION STEPS — §7.2

[M] Mila should organize a Canadian AI labour displacement scenario exercise with AI researchers, labour economists, ESDC officials, TechStat analysts, and provincial labour market representatives to evaluate new and existing policy responses across a set of plausible AI displacement scenarios.

[M] Mila should produce a working document from the exercise that maps plausible AI displacement scenarios to candidate policy responses and gives ESDC a starting point for designing the specific interventions Canada would need to deploy.

[J] TechStat and ESDC's Labour Market Information Directorate should establish working sessions on how to interpret the early warning system's outputs, including what patterns count as meaningful warning signals and at what point those signals warrant escalation.

Actor key: **[M]** Mila **[T]** TechStat **[E]** ESDC **[J]** Joint TechStat + ESDC

PART 3

Recommendations

CHAPTER 08

Recommendations and Implementation Plan

CHAPTER AT A GLANCE

This chapter consolidates the action items shared throughout the report into six distinct recommendations. Three are directed towards Mila and three are for the Government of Canada. The recommendations are followed by an implementation plan that identifies the specific timelines, stakeholders, and venues needed for each recommendation.

8.1 Recommendations for Mila

RECOMMENDATION 1

Brief TechStat on the indicator framework in this report and advocate for the data streams it will need, engaging in Q2 2026 before TechStat's first-year work plan is finalized.

ACTION STEPS

[M] Brief TechStat on the indicator framework including the automation-augmentation distinction, value of high-frequency payroll data, and the limits of static exposure indices like C-AIOE compared to dynamic task exposure metrics. (§5)

[M] Brief TechStat on why the early warning system should be designed to operate on domestic data streams first, with frontier lab telemetry integrated as a plug-in upgrade when available, so that the system's functionality does not depend on international standardization timelines. (§5.1, §6.3)

[M] Flag CSBC discontinuation to TechStat as a near-term data risk, and support TechStat through ISED if needed. (§6.1.2)

[M] Brief TechStat on payroll partnership technical and legal considerations. (§6.2)

[M] Advise TechStat on Phoenix data limitations as a source of public sector payroll data. (§6.2.1)

[M] Brief TechStat on COVID-19 case study findings, particularly the employment verification gap that produced the CERB overpayment problem. (§7.1)

RECOMMENDATION 2

Convene a Canadian AI labour displacement scenario exercise in October or November 2026 that produces a working document mapping plausible scenarios to candidate policy responses.

ACTION STEPS

[M] Organize a Canadian AI labour displacement scenario exercise with AI researchers, labour economists, ESDC officials, TechStat analysts, and provincial labour market representatives to evaluate policy responses (new and existing) across a set of plausible AI displacement scenarios. (§7.2)

[M] Produce a working document from the exercise that maps plausible AI displacement scenarios to candidate policy responses and gives ESDC a starting point for designing the specific interventions Canada would need to deploy. (§7.2)

RECOMMENDATION 3

Consult provinces through bilateral outreach to Ontario and Quebec and a joint technical briefing with the Labour Market Information Council and communicate Canada's data requirements to the international AI usage data standard effort.

ACTION STEPS

[M] Engage provincial labour market officials to ensure they understand what the early warning system will produce and what kinds of decisions they can inform. (§5.5)

[M] Consult with provincial labour market ministries through Forum of Labour Market Ministers (FLMM) to understand their data needs, including the appropriate level of geographic disaggregation, and communicate Canada's data requirements with Manning's team. (§6.3)

8.2 Recommendations for the Government of Canada

RECOMMENDATION 4

TechStat should secure three domestic data streams in 2026: an LFS digital platform supplement at quarterly frequency, absorption of the CSBC AI adoption module before its August 2026 sunset, and voluntary data-sharing partnerships with ADP Canada and Dayforce.

ACTION STEPS

[T] TechStat should use the LFS as an interim data stream and request that the LFS digital platform supplement move from annual to quarterly frequency. (§6.1.1)

[T] TechStat should absorb the CSBC's AI adoption module before it sunsets in August 2026. (§6.1.2)

[T] TechStat should secure voluntary data-sharing partnerships with ADP Canada and Dayforce with the goal of beginning monthly data deliveries by early 2027. (§6.2)

[T] TechStat should assess whether a workable translation pathway between U.S. and Canadian occupational classification systems will be in place before international AI usage data arrives. (§6.3)

RECOMMENDATION 5

TechStat and ESDC's Labour Market Information Directorate should begin quarterly calibration sessions in mid-2026 to develop shared signal interpretation, escalation thresholds, and signal-to-response mapping.

ACTION STEPS

[J] TechStat and ESDC's Labour Market Information Directorate should establish working sessions on how to interpret the early warning system's outputs, including what patterns count as meaningful warning signals and at what point those signals warrant escalation. (§7.2)

[E] ESDC should use the working document from Mila's scenario exercise (Recommendation 2) as an input to start identifying likely AI displacement benefit designs and assessing their implementation requirements. (§7.1 and §7.2)

RECOMMENDATION 6

ESDC should table two sequenced items at the FLMM's Changing Nature of Work and Skills Working Group: an awareness agenda item in late 2026 and a structured working session on means-testing treatment of future AI displacement benefits in mid-2027.

ACTION STEPS

[E] ESDC should use the Forum of Labour Market Ministers' Changing Nature of Work and Skills Working Group to establish proactive agreements with provinces on how a future federal AI displacement benefit would be treated under provincial means-testing rules. (§7.1)

Actor key: **[M]** Mila **[T]** TechStat **[E]** ESDC **[J]** Joint TechStat + ESDC

8.3 Implementation Plan

Note: Appendix F identifies the specific individuals whose decision-making authority is relevant to each recommendation.

RECOMMENDATION 1

Brief TechStat on the indicator framework in this report and advocate for the data streams it will need, engaging in Q2 2026 before TechStat's first-year work plan is finalized.

Format: Mila should prepare a written briefing document that TechStat officials can circulate internally after the session. The six action steps should be organized into the four blocks below and presented in a single half-day working session rather than a series of separate meetings, both for calendar efficiency and because the briefings reinforce each other. The recommended sequence is:

1. Indicator framework (§5), since it shapes how every subsequent briefing is received
2. The CSBC discontinuation risk (§6.1.2) due to its summer 2026 sunset, the LFS supplement request (§6.1.1), and the domestic-data-first design principle (§5.1, §6.3)
3. Payroll partnership technical and legal considerations (§6.2), including the limits of the Phoenix pay system as a source of public sector payroll data (§6.2.1)
4. Lessons from the COVID-19 emergency response, particularly the employment verification gap that produced the CERB overpayment problem (§7.1)

What should come out of it: The briefing succeeds if TechStat agrees to three things: incorporating the CSBC AI module absorption into its immediate priorities before the August 2026 sunset, initiating outreach to ADP Canada and Dayforce for voluntary payroll partnerships, and adopting a "domestic-first" architecture so that the system's functionality does not depend on the international lab standardization timeline.

If the window closes: If TechStat's first-year work plan is finalized before the briefing takes place, the CSBC absorption becomes the most urgent standalone item because its August 2026 sunset is a hard deadline.

RECOMMENDATION 2

Convene a Canadian AI labour displacement scenario exercise in October or November 2026 that produces a working document mapping plausible scenarios to candidate policy responses.

Mila should host the scenario exercise in October or November 2026 to avoid December's drop in public-sector engagement, with the working document produced by year-end.

Format: The exercise should follow the structured scenario model used by Windfall Trust in its February 2026 London convening, where participants worked in breakout groups to produce specific policy recommendations under a defined scenario. A full-day working session with breakout groups produces a more actionable output than a panel discussion or expert survey, because participants can draft and pressure-test policy responses in real time.

Participants should include:

- AI researchers from Mila and participants in Mila's AI Policy Fellowship (workforce reskilling thematic area).
- International researchers driving the AI displacement research forward, such as Erik Brynjolfsson (Stanford Digital Economy Lab), David Autor (MIT), Ajay Agrawal (Rotman, University of Toronto), Sam Manning (Centre for the Governance of AI), and Anton Korinek (Centre for the Governance of AI). Ketana Krishna (Oxford Martin AI Governance Initiative) would also be a valuable participant given her direct involvement in the international lab standardization discussions covered in Chapter 6.

- Labour economists from institutions like the Dais at Toronto Metropolitan University, Institute for Research on Public Policy (IRPP), Université de Montréal, CIRANO, and the Labour Market Information Council (LMIC).
- Members of ESDC's Labour Market Information Directorate which translates labour market signals into program responses, TechStat analysts, and provincial labour market representatives from at least Quebec and Ontario.

What the working document should contain: The exercise should produce a scenario-to-policy matrix with columns for speed of disruption, candidate policy instruments, delivery channel (EI, CRA, or new), legislative requirements (if any), estimated implementation timeline, and federal-provincial coordination requirements. The document should be formatted as a working input for ESDC, which means it should identify trade-offs and implementation constraints rather than advocating for a single preferred option. Mila should plan the document's format and distribution with this handoff in mind and deliver the final version to both TechStat and ESDC's Labour Market Information Directorate by January 2027.

How this feeds into Recommendation 5: The scenario-to-policy matrix is a preparatory document which answers the question: if AI displacement unfolds at a given speed, in a given set of occupations, at a given severity, what policy instruments could Canada deploy? The matrix exists before any real data arrives and provides a menu of candidate responses organized by scenario type. This provides TechStat and ESDC with a set of pre-identified candidate responses which they can map signals onto when building their response protocol (Recommendation 5, Phase 3).

RECOMMENDATION 3

Consult provinces through bilateral outreach to Ontario and Quebec and a joint technical briefing with the Labour Market Information Council and communicate Canada's data requirements to the international AI usage data standard effort.

The provincial consultations should begin in Q2 2026, well before the arrival of AI usage statistics, and continue through year-end . Mila's engagement should combine two channels:

1. **Bilateral outreach to Ontario and Quebec:** Ontario accounts for an estimated 43.4% of young "canary" workers, and Quebec is Mila's home jurisdiction and where it has the

strongest existing relationship base. The right counterparts are the senior officials responsible for labour market policy and labour market information within Ontario's Ministry of Labour, Immigration, Training and Skills Development and Quebec's Ministère de l'Emploi et de la Solidarité sociale.[93] Mila should aim for two or three substantive conversations per province. The goal of these conversations is to collect specific requirements: what level of geographic and occupational disaggregation do provinces need to act on the early warning system's signals, what decision-making processes would the signals feed into, and what format makes the outputs usable for provincial officials who are not data scientists.

2. **Technical briefing through the Labour Market Information Council (LMIC):** LMIC is the Forum of Labour Market Ministers' (FLMM) research arm and has standing relationships with labour market information officials in all 13 provinces and territories, plus ESDC and Statistics Canada. Mila should approach LMIC (Executive Director Rebecca Webb, Board Chair Suzanne Harbottle) and propose a joint briefing on the early warning system's data requirements.[94] The purpose of the LMIC briefing is to build awareness among provinces and territories that have not been part of the bilateral conversations, and to surface any data requirements from smaller jurisdictions that Ontario and Quebec may not have shared. This briefing should take place after the bilateral conversations with Ontario and Quebec have produced initial findings, so that Mila can present a draft data specification rather than an open-ended consultation.

RECOMMENDATION 4

[93] Ministère de l'Emploi et de la Solidarité sociale, homepage, Gouvernement du Québec, last updated February 12, 2026, accessed April 12, 2026, https://www.quebec.ca/gouvernement/ministeres-organismes/emploi-solidarite-sociale; Ministry of Labour, Immigration, Training and Skills Development, homepage, Government of Ontario, last updated August 12, 2025, accessed April 12, 2026, https://www.ontario.ca/page/ministry-labour-immigration-training-skills-development.

[94] Newswire, "Labour Market Information Council (LMIC) Appoints Rebecca Webb as Executive Director to Advance Pan-Canadian Labour Market Collaboration," news release, May 21, 2025, https://www.newswire.ca/news-releases/labour-market-information-council-lmic-appoints-rebecca-webb-as-executive-director-to-advance-pan-canadian-labour-market-collaboration-814712260.html.

TechStat should secure three domestic data streams in 2026: an LFS digital platform supplement at quarterly frequency, absorption of the CSBC AI adoption module before its August 2026 sunset, and voluntary data-sharing partnerships with ADP Canada and Dayforce.

LFS supplement request and approval: The LFS supplement request should be made directly to Josée Bégin's directorate, the Assistant Chief Statistician who oversees the Social, Health and Labour Statistics at StatsCan. The request should specify the digital platform supplement as a quarterly addition to the existing LFS instrument. TechStat should prepare a short justification memo explaining why quarterly frequency is necessary.

CSBC absorption: The CSBC absorption work is the most time-sensitive and should be initiated immediately before the survey sunsets in August 2026.

Payroll data-sharing agreement negotiations:

Disclaimer: *The author is not a lawyer, and the legal interpretations in this report are the product of independent research into relevant primary statutory sources. This is not legal advice. TechStat, Statistics Canada, the payroll processors, and the Office of the Privacy Commissioner must all consult qualified legal counsel before any partnership is negotiated or signed.*

Payroll partnerships should be initiated with both ADP Canada and Dayforce simultaneously in early 2026 with the target of receiving the monthly data deliveries by early 2027. The partnerships should be structured as voluntary data-sharing agreements under PIPEDA Section 7(3)(f).[95] Section 17 of the Statistics Act provides the confidentiality shield once the data enters Statistics Canada.[96]

The steps are as follows:

- Initial outreach to key decision-makers at each processor (Jim Lord, President of ADP Canada; Dr. Nela Richardson, ADP Chief Economist; David Ossip, CEO at Dayforce). Reference the existing ADP-Stanford Digital Economy Lab partnership as a model for a voluntary partnership. *(Q2 2026, approximately 2 to 3 months for initial conversations and assessment of willingness.)*

[95] Personal Information Protection and Electronic Documents Act, S.C. 2000, c. 5, s. 7(3)(f), https://laws-lois.justice.gc.ca/eng/acts/p-8.6/FullText.html.
[96] Statistics Act, R.S.C. 1985, c. S-19, s. 17, https://laws-lois.justice.gc.ca/eng/acts/s-19/FullText.html.

- Necessity and proportionality assessment prepared by Statistics Canada following the framework adopted after the 2018 TransUnion investigation. *(Can proceed in parallel with initial outreach, approximately 2 to 3 months.)*
- Privacy Impact Assessment led by Statistics Canada's Office of Privacy Management and Information Coordination. The PIA should address Quebec's Law 25 obligations that apply to any Quebec-based payroll processor transferring data outside the province.[97] *(Approximately 2 to 3 months, partially overlapping with the necessity and proportionality assessment.)*
- Pre-consultation with the Office of the Privacy Commissioner along with the payroll companies, presenting the PIA and necessity and proportionality assessment. This isn't strictly necessary but recommended given the TransUnion precedent. *(Approximately 1 to 2 months.)*
- Negotiation and signing of the Administrative Data Acquisition Agreement, specifying legal authority, data fields, aggregation thresholds, delivery cadence, termination provisions, and Section 17 confidentiality commitments. *(Approximately 2 to 3 months.)*
- Pilot delivery and validation, followed by a transition to recurring monthly or bi-weekly delivery. *(Approximately 1 to 2 months.)*

The total elapsed time from initial outreach to first recurring data delivery is approximately 8 to 13 months if steps are sequenced with partial overlap where possible. This places the realistic target for first data deliveries between Q1 and Q2 2027.

A detailed walkthrough of the relevant legal provisions (PIPEDA Section 7(3)(f), Statistics Act Section 17, the Administrative Data Acquisition Agreement vehicle, the Necessity and Proportionality Framework, and Quebec's Law 25) is provided in **Appendix D**.

[97] OneTrust, "Quebec's Law 25: What Is It and What Do You Need to Know?" OneTrust (blog), accessed April 12, 2026, https://www.onetrust.com/blog/quebecs-law-25-what-is-it-and-what-do-you-need-to-know/

RECOMMENDATION 5

TechStat and ESDC's Labour Market Information Directorate should begin quarterly calibration sessions in mid-2026 to develop shared signal interpretation, escalation thresholds, and signal-to-response mapping.

TechStat's counterpart within ESDC is the Labour Market Information Directorate within the Skills and Employment Branch, identified in Chapter 4 as the unit that produces labour market intelligence for policy use and coordinates with Statistics Canada on survey products.[98]

The working sessions between TechStat and ESDC should begin in mid-2026 and continue on a quarterly basis thereafter.

The working sessions should address three components, to be completed sequentially rather than simultaneously:

1. ***Phase 1 (mid-2026 onward) – Signal interpretation:*** The first sessions should focus entirely on developing shared definitions of what patterns in the data constitute meaningful early warning signals versus normal variation. This work can begin as soon as TechStat has any data flowing, including LFS data, CSBC AI adoption data (if the module is successfully absorbed), and early pilot deliveries from payroll partnerships. The two teams should work through historical and simulated data together to build a shared analytical vocabulary before live signals arrive at scale.
2. ***Phase 2 (late-2026 to early 2027) – Escalation thresholds:*** Once TechStat and ESDC have enough shared experience interpreting data to distinguish meaningful patterns from noise, the sessions should shift to defining escalation thresholds. The calibration sessions are where the two teams develop the criteria for three tiers of escalation:
    a. **Tier 1:** Signals that trigger an internal ESDC review within the Labour Market Information Directorate, actionable on the Directorate's own authority.
    b. **Tier 2:** Signals that warrant a brief to the Minister of Jobs and Families (Patty Hajdu). The decision to escalate to this tier should rest with ESDC's Senior Assistant Deputy

[98] Employment and Social Development Canada, "About Us."

Minister for the Skills and Employment Branch (Colette Kaminsky), based on the criteria developed in the calibration sessions.

c. **Tier 3:** Signals that warrant convening the provinces through the FLMM. This is a significant political action that requires ministerial authorization. The calibration sessions define what would justify this step and the Minister (Patty Hajdu) decides whether to take it.

3. ***Phase 3 (early 2027 onward) – signal-to-response mapping:*** This component begins after the working document from Mila's scenario-to-policy mapping exercise (Recommendation 2) is delivered, expected by January 2027. Where the scenario-to-policy matrix is preparatory and hypothetical (if displacement unfolds at this speed and severity, here are the candidate responses), signal-to-response mapping connects specific patterns in TechStat's data to specific entries in the matrix. For example, if payroll data shows a 12% quarter-over-quarter headcount decline in customer service occupations concentrated in two census metropolitan areas (CMAs), and AI lab usage statistics show a simultaneous spike in automation-mode usage for customer service tasks, the signal-to-response protocol tells TechStat and ESDC which escalation tier that pattern triggers and which row of the scenario-to-policy matrix contains the candidate response. The calibration sessions should work through a set of simulated signals against the matrix to pressure-test the mapping.

ESDC's internal policy design work: The calibration sessions will inform but cannot substitute for a separate policy development exercise within ESDC. Designing AI displacement benefit parameters, assessing delivery channel options (EI, CRA, or a new instrument), estimating implementation timelines, and defining eligibility criteria are dedicated policy design tasks that ESDC owns independently. ESDC should assign this work to a team within the Skills and Employment Branch that uses the Recommendation 2 working document as its starting input and the calibration sessions' signal-to-response mapping as an ongoing feedback loop. The calibration sessions surface what signals look like and when they cross escalation thresholds. ESDC's policy design work determines how the government responds.

RECOMMENDATION 6

ESDC should table two sequenced items at the FLMM's Changing Nature of Work and Skills Working Group: an awareness agenda item in late 2026 and a structured working session on means-testing treatment of future AI displacement benefits in mid-2027.

ESDC should engage the provinces through the Forum of Labour Market Ministers (FLMM), which is co-chaired for the 2025-2027 cycle by the federal Minister of Jobs and Families, Patty Hajdu, who also leads ESDC.[99] The appropriate venue for this is the FLMM's **Changing Nature of Work and Skills Working Group (CNWSWG)**, whose mandate is to "support a strong and responsive labour market that is well prepared for labour market disruptions."[100]

ESDC and TechStat should table two related items at the CNWSWG:

1. **An AI labour displacement agenda item in late 2026.** ESDC and TechStat should present three things to the CNWSWG: what TechStat is and how it fits within Statistics Canada's mandate, what data streams it is acquiring (LFS supplement, CSBC AI module, payroll partnership pilot data if available by that point), and what kinds of early warning signals it will produce for provincial labour market officials. If payroll pilot deliveries have begun under the Recommendation 4 timeline, ESDC should present preliminary outputs to give provincial officials a sense of what the system looks like in practice. This session helps the ministers and senior officials understand the system and the kinds of signals it will produce before they are asked to discuss benefit coordination in the second session.

2. **A structured working session focused on the means-testing treatment of one or more future AI displacement benefit(s) in mid-2027.** The goal is to establish proactive agreements with provinces on how a future federal AI displacement benefit would be treated under provincial means-testing rules, so that the claw-back problem documented in §7.1 (where CERB payments triggered automatic reductions to provincial social assistance in several provinces) does not repeat. For the sessions to be productive, provinces need to know what policy solutions and benefit designs are being proposed before they can agree on how their systems should treat them. The working session should

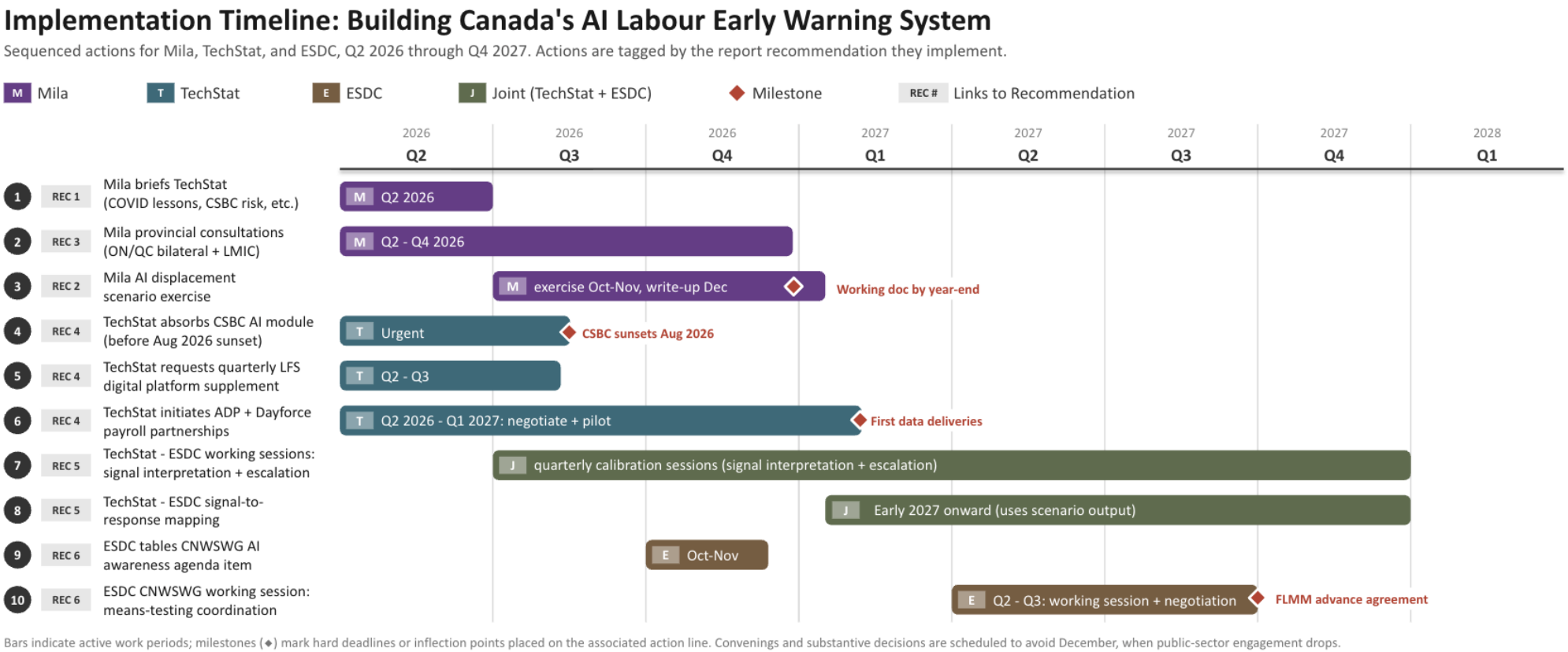
[99] Forum of Labour Market Ministers, "FLMM and Labour Market News," FLMM-FMMT, accessed April 12, 2026, https://flmm-fmmt.ca/flmm-and-labour-market-news/.
[100] Forum of Labour Market Ministers, Annual Report 2021-2022, FLMM-FMMT, https://flmm-fmmt.ca/wp-content/uploads/2025/03/FLMM-2021-22-Annual-Report-External-Final.pdf.

draw on the candidate policy solutions identified in the scenario-to-policy matrix from Mila's convening (Recommendation 2, delivered by January 2027) and refined through ESDC's internal policy design work (Recommendation 5). ESDC should present one or two specific candidate benefit designs and discuss exemptions from provincial means-testing.

Sequencing relative to other recommendations: This sequencing ensures that ministers and senior officials understand the system and the kinds of signals it will produce before they are asked to discuss advance agreements on benefit coordination. If the Recommendation 2 working document is delayed past January 2027, ESDC should consider postponing the mid-2027 working session rather than tabling it without specific proposals for provinces to respond to.

8.4 Implementation Timeline

This timeline consolidates the eleven action steps from the six recommendations in Chapter 8 into a single sequenced view spanning Q2 2026 through Q4 2027. Each action is tagged by the recommendation it implements (REC 1 through REC 6) and color-coded by lead actor.



APPENDICES

Supporting material, methodology, and reference documents

APPENDIX A

Canary Cohort Crosswalk Methodology

Source definition

The "canary cohort" is derived from "Canaries in the Coal Mine?" by Brynjolfsson et al. which correlated employment declines associated with generative AI adoption in workers aged 22 to 25 in highly AI-exposed occupations where AI was being used to automate tasks.[101]

Crosswalk to Canadian classifications

The U.S. occupations were translated to Canada's National Occupational Classification (NOC) 2021: SOC 2018 (US) → NOC 2016 → NOC 2021. O*NET occupations are built on the SOC 2018 framework, so I started from the SOC 2018 instead of O*NET.[102] Both crosswalk steps are published by Statistics Canada. Applying both correspondence steps yields seven NOC 2021 unit groups.

Source U.S. occupation	SOC 2018	NOC 2021	Occupation name
Software Developers	15-1252	**21231**	Software engineers and designers
Software Developers	15-1252	**21232**	Software developers and programmers
Software Developers	15-1252	**21234**	Web developers and programmers
Customer Service Representatives	43-4051	**64409**	Other customer and information services representatives
Tellers (bank tellers)	43-3071	**64400**	Customer services representatives, financial institutions

[101] Brynjolfsson, Chandar, and Chen, "Canaries in the Coal Mine?"
[102] Statistics Canada, "SOC 2018-NOC 2016 Concordance;" Statistics Canada, "NOC 2016–NOC 2021 Concordance."

Travel Agents	41-3041	**64310**	Travel counsellors
Tax Preparers	13-2082	**14200**	Accounting and related clerks

Statistics Canada extraction

Employment data were retrieved from Statistics Canada's Table 98-10-0593-01.[103] Dimension selections: Occupation (the seven NOC codes above), Statistics (Number), Labour force status (Employed), Age (20 to 24 years), Gender (Total), Class of worker (Total), Geography (Canada).

Cohort sizing results

NOC 2021	Occupation	Total, all ages	Aged 20-24	Youth share
64409	Other customer and information services reps	181,700	29,605	16.3%
21232	Software developers and programmers	155,700	8,005	5.1%
14200	Accounting and related clerks	151,100	8,815	5.8%
21231	Software engineers and designers	113,100	5,805	5.1%
64400	Customer services reps, financial institutions	71,500	9,400	13.1%
21234	Web developers and programmers	28,700	4,770	16.6%
64310	Travel counsellors	14,600	305	2.1%

[103] Statistics Canada, "Table 98-10-0593-01: Class of Worker by Occupation Unit Group, Labour Force Status, Age and Gender: Canada, Provinces and Territories, Census Metropolitan Areas and Census Agglomerations with Parts," 2021 Census of Population, accessed April 12, 2026, https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=9810059301.

Total	**All seven canary occupations**	**716,400**	**66,705**	**9.3%**

Methodological caveats

- **Age bracket proxy:** The Brynjolfsson paper defines the canary group as workers aged 22 to 25. The closest available bracket in Table 98-10-0593-01 is 20 to 24, which I used as the closest comparison.
- **Reference period:** Census data refer to the week of May 2 to 8, 2021, during COVID-19 recovery. Travel counsellors and customer service occupations may have been temporarily depressed compared to baseline.
- **Geographic distribution:** The provincial figures cited in Chapter 3.2.1 are extracted directly from Table 98-10-0593-01, filtered to employed persons aged 20 to 24 in the seven canary NOC codes. Provincial totals sum to 66,695, a 10-person discrepancy from the national total of 66,705 due to Statistics Canada's standard practice of rounding individual cells to the nearest 5 for confidentiality.

APPENDIX B

Limits of the Canadian Statistical Apparatus

Early manifestations of AI-driven labour disruption are localized within specific micro-cohorts that fall below the detection threshold of Canada's existing government surveys:

Labour Force Survey (LFS): Canada's official monthly employment baseline is built on a rotating sample of approximately 65,000 households. The LFS is the only source of monthly employment estimates for the entire Canadian labour force and is the most representative labour market instrument Statistics Canada operates.[104] However, once the sample is sliced by occupation, age, and province, cell sizes become too small to be statistically significant.

Survey of Employment, Payrolls and Hours (SEPH): SEPH provides a near-census view of jobs, earnings, and hours by drawing on payroll deduction filings from employers across Canada.[105] It is significantly larger in scope than the LFS but is built around industry classification rather than occupational classification. SEPH cannot produce the occupation-by-age breakdowns required to detect AI-driven displacement at the cohort level identified in this report.

The Census of Population: The Census provides the exact occupational, demographic, and geographic depth required to identify cohorts like the canary group quantified in Chapter 3, including the Statistics Canada Table 98-10-0593-01 used in **Appendix A**.[106] However, the Census is conducted on a five-year cycle, which makes it unusable as a real-time monitoring instrument.

[104]Statistics Canada, "Labour Force Survey."
[105]Statistics Canada, "Survey of Employment, Payrolls and Hours (SEPH)," Survey No. 2612, accessed April 2026, https://www23.statcan.gc.ca/imdb-bmdi/pub/2612-eng.htm.
[106]Statistics Canada, "Chapter 1: Introduction," Guide to the Census of Population, 2021, Catalogue no. 98-304-X (Ottawa: Statistics Canada, 2021), https://www12.statcan.gc.ca/census-recensement/2021/ref/98-304/2021001/chap1-eng.cfm.

APPENDIX C

Populations Excluded from Commercial Payroll Coverage

As discussed in Chapter 5.3.2, commercial payroll data covers an estimated 80% of the Canadian workforce. The remaining 20% consists of populations that fall outside enterprise payroll platforms, primarily because they are not captured by the T4 reporting system that commercial payroll providers process.[107]

Self-employed workers and gig workers: Approximately 2.7 million Canadians are self-employed, representing roughly 13% of the workforce, and an additional 1.7 million (around 8%) engage in some form of gig work.[108] Approximately 75% of gig workers simultaneously hold a traditional T4 job, which means they would still appear in commercial payroll data through their employer-of-record relationship.[109]

Micro-businesses: Businesses with fewer than five employees rarely use enterprise payroll platforms, instead using platforms like Quickbooks Payroll or Gusto. These businesses represent roughly 6% of the total payroll workforce.[110]

Implications for monitoring: For gig workers specifically, the Labour Force Survey could serve as a partial monitoring tool because the cohort-level granularity required for the canary population may not apply to the same degree. For example, generative AI may reduce demand for freelance copywriting because clients use AI tools instead, and this disruption would plausibly affect freelance copywriters similarly regardless of age band. However, even setting aside age as a filter, the number of AI-exposed gig workers within the LFS sample may still be too small given the smaller base rate to produce statistically significant estimates at the occupation level.

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[107] Staffing Industry Analysts, "Canada's Self-Employed at 13.2%;" Statistics Canada, "Labour Statistics at a Glance: Experiences of Self-Employed Workers in Canada, 2023," Catalogue no. 71-222-X (Ottawa: Statistics Canada, 2024), https://www150.statcan.gc.ca/n1/pub/71-222-x/71-222-x2024001-eng.htm.

[108] Statistics Canada, "Labour Statistics at a Glance: Experiences of Self-Employed Workers in Canada, 2023," Catalogue no. 71-222-X (Ottawa: Statistics Canada, 2024), https://www150.statcan.gc.ca/n1/pub/71-222-x/71-222-x2024001-eng.htm.

[109] Canadian HR Reporter, "How Many Canadians Are Gig Workers?," October 9, 2024, https://www.hrreporter.com/focus-areas/compensation-and-benefits/how-many-canadians-are-gig-workers/389027.

[110] Statistics Canada, "Table 14-10-0215-01."

The dynamics of AI displacement among self-employed workers and micro-businesses were not investigated in depth by this report. These are acknowledged limitations of the early warning system as currently designed. The recommendation to anchor monitoring on commercial payroll data remains sound for the approximately 80% of the workforce it covers.

APPENDIX D

Payroll Data Agreement: Legal and Procedural Steps

A note on scope and a disclaimer

The voluntary payroll partnerships proposed in Chapter 6.2 are first-of-their-kind in Canada. No Canadian payroll processor has previously shared payroll data with Statistics Canada under any legal instrument, and no published template exists for how such a partnership would be structured. This appendix is the product of the author's own research into the primary statutory sources (PIPEDA, the Statistics Act, and related regulations), the Office of the Privacy Commissioner's published findings, and Statistics Canada's public policy framework.

The author is not a lawyer. Nothing in this appendix constitutes legal advice. The purpose of the appendix is to surface the legal questions that TechStat will need to resolve, identify the statutory provisions that appear to govern the partnership, and flag the key precedents and guardrails that should inform its design. TechStat, Statistics Canada, the payroll processors, and the Office of the Privacy Commissioner must all consult with qualified legal counsel before any partnership is negotiated or signed. The statutory interpretations offered below are starting points for legal review, not final conclusions.

Legal architecture: three provisions in combination

The legal pathway for a voluntary payroll partnership appears to rest on the combination of three existing statutory provisions.

PIPEDA Section 7(3)(f): the statistical-purposes disclosure exemption

What it says: Section 7(3) of the Personal Information Protection and Electronic Documents Act opens with the general rule that governs disclosure of personal information by private-sector organizations:

7 *(3)* *For the purpose of clause 4.3 of Schedule 1, and despite the note that accompanies that clause, an organization may disclose personal information without the knowledge or consent of the individual only if the disclosure is*

The subsection then lists the permitted exceptions. Paragraph (f) reads:

- ***(f)*** *for statistical, or scholarly study or research, purposes that cannot be achieved without disclosing the information, it is impracticable to obtain consent and the organization informs the Commissioner of the disclosure before the information is disclosed;*[111]

Possible implication: This provision is the most direct statutory basis in Canadian law for a commercial payroll processor to share employee payroll data with Statistics Canada for labour market statistical purposes without first obtaining consent from every employee whose data would be included. A processor like ADP Canada or Dayforce handles records for millions of employees across thousands of employer clients, and obtaining individual consent from each would be operationally impossible. The statistical purpose (tracking labour market changes associated with AI adoption) is within the plain meaning of the provision. The Privacy Commissioner notification requirement is a procedural step, though in practice the Commissioner's views would heavily shape how the partnership is designed.

Statistics Act Section 17

What it says: Section 17(1) of the Statistics Act reads:

17 (1) *Except for the purpose of communicating information in accordance with any conditions of an agreement made under section 11 or 12 and except for the purposes of a prosecution under this Act but subject to this section,*

- ***(a)*** *no person, other than a person employed or deemed to be employed under this Act, and sworn under section 6, shall be permitted to examine any return or identifying information obtained for the purposes of this Act; and*
- ***(b)*** *no person who has been sworn under section 6 shall disclose or knowingly cause to be disclosed, by any means, any information obtained under this Act in a manner that it is possible from the disclosure to relate the information obtained to any identifiable individual person, business or organization.*[112]

Author's interpretation: Once data enters Statistics Canada under any collection authority, only sworn Statistics Canada employees (or "deemed employees," meaning contractors who have taken the formal oath of secrecy) can examine the data in identifiable form. Then even sworn

[111] PIPEDA.
[112] Statistics Act, s. 17.

employees cannot disclose the data outside the agency in any form that would make it possible to identify a specific person, business, or organization. Section 17 does not prohibit all external disclosure, but only disclosure in a manner that can lead to the identification of a person, business, or organization. Violating Section 17 is a criminal offence under Sections 30 and 31 of the Statistics Act.[113]

Possible implication: Section 17 is the strongest confidentiality guarantee in Canadian federal statistical law. Its practical effect for a payroll processor contemplating a partnership is that once the data has been transferred to Statistics Canada, it cannot be used by any other federal department for administrative, regulatory, enforcement, or tax purposes. The CRA cannot request the underlying records for audit purposes. ESDC cannot pull individual records for benefits adjudication. This can be a point of assurance for payroll processors and their clients who would reasonably fear sharing their employee data with a federal body.

TechStat can and must produce statistical products (dashboards, trend indicators, early warning signals) that are then shared with ESDC, the FLMM, the Prime Minister's Office, provincial labour ministries, and the public. A TechStat dashboard shared with ESDC that displays, "employment in NOC 21232 (software developers) in the Toronto census metropolitan area (CMA) fell 8% between January and June 2027" would likely not violate Section 17.

Lastly, if an ESDC analyst (or a provincial labour ministry analyst, or an academic researcher) wants to work directly with the identifiable payroll microdata rather than with TechStat's vetted outputs, Section 17 seems to require them to become a "deemed employee" of Statistics Canada under Section 5 of the Act, and take the formal oath of secrecy:

- ***5 (1)*** *The Chief Statistician may employ, in the manner authorized by law, any commissioners, enumerators, agents or other persons that are necessary to collect for Statistics Canada the statistics and information that the Minister considers useful and in the public interest relating to the commercial, industrial, financial, social, economic and other activities that the Minister may determine. The duties of the commissioners, enumerators, agents or other persons shall be those duties prescribed by the Chief Statistician.*[114]

[113] Ibid.
[114] Ibid.

The Administrative Data Acquisition Agreement (ADAA)

What it is: The Administrative Data Acquisition Agreement is the standard contractual instrument Statistics Canada uses to receive data from external providers, whether those providers are other federal departments, provincial governments, or private-sector organizations. A typical ADAA specifies Statistics Canada's legal authority to receive the data, the intended statistical use, the data fields being transferred, the delivery cadence, security and confidentiality requirements, and termination provisions.[115]

Existing precedent: Statistics Canada's longest-running voluntary private sector data-sharing arrangements are its partnerships with Canadian grocery retailers to receive point-of-sale scanner data.

Why Section 13 compulsion is off the table

Statistic Canada's 2017-2018 attempt to compel financial transaction data from banks and credit bureaus under Section 13 triggered 103 complaints to the Privacy Commissioner, parliamentary hearings, and the suspension of both projects following sustained public and political backlash. The OPC's December 2019 investigation judged that the projects were legally authorized but failed the necessity and proportionality test.[116] The OPC recommended a "four-part test" before any new data acquisition:

- *"Is the measure demonstrably necessary to meet a specific need?"*
- *"Is it likely to be effective in meeting that need?"*
- *"Is the loss of privacy proportional to the need?"*
- *"Is there a less privacy-invasive way of achieving the same end?"*[117]

[115] Statistics Canada, "Policy on the Use of Administrative Data Obtained under the Statistics Act," app. A ("Definitions"), accessed April 12, 2026, https://www.statcan.gc.ca/en/about/transparency/policy/administrative-data.
[116] OPC, "Investigation into Statistics Canada's Collection of Financial Transactions Information."

[117] Ibid.

Statistics Canada subsequently adopted a Necessity and Proportionality Framework requiring a four-part test before any new data acquisition.[118]

Implication: The OPC's guidance strongly points toward voluntary, aggregated, necessary / proportionate data collection and against compulsory actions.

Provincial Legislation

Quebec's Law 25 (formally, An Act to modernize legislative provisions as regards the protection of personal information, phased in between 2022 and 2024) is the strictest privacy regime in Canada and imposes GDPR-style requirements that include mandatory privacy impact assessments before any transfer of personal information outside Quebec, data portability rights, and administrative penalties of up to $10 million.[119] For the payroll partnership, Law 25 would likely sets the standard for the ADAA design, but legal counsel should still confirm this assessment.

The Pre-Aggregation Alternative

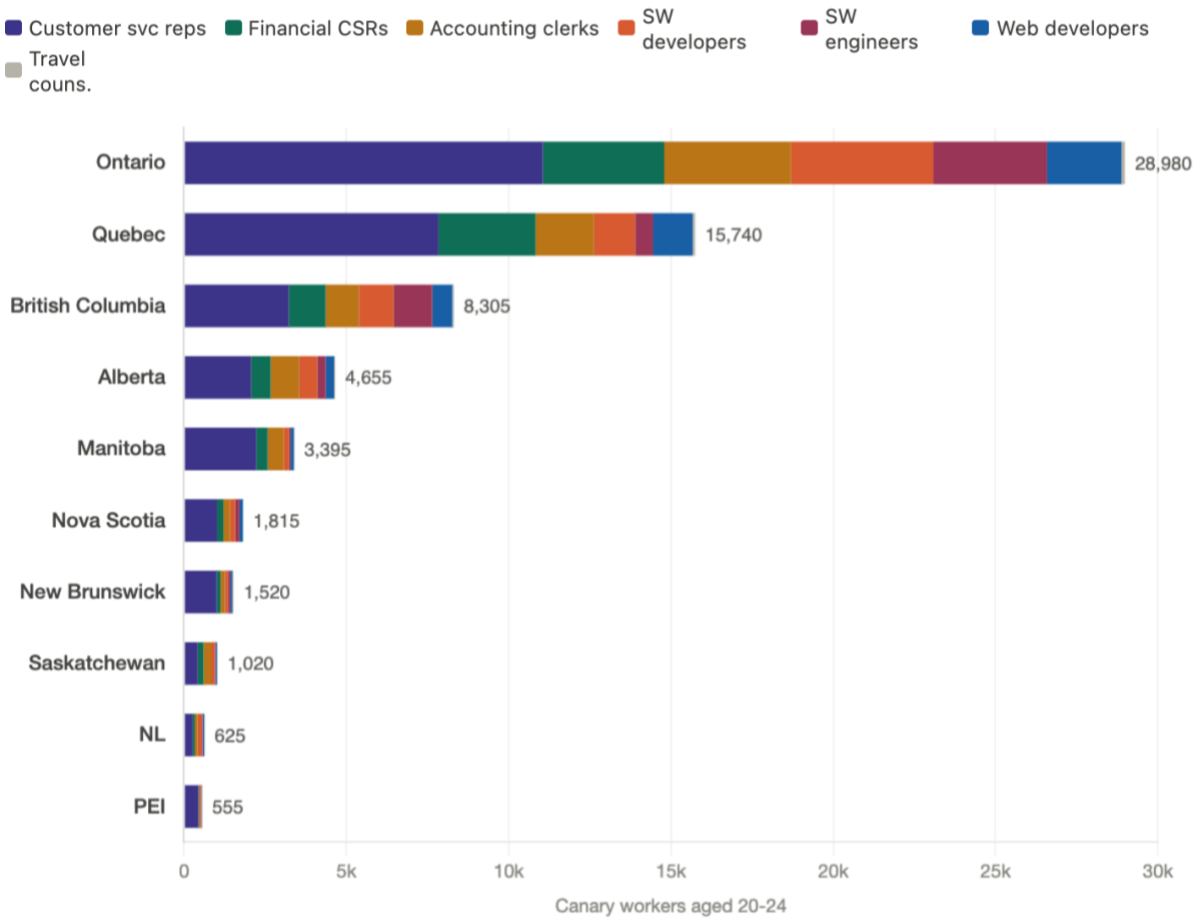
Rather than transferring individual-level payroll records to Statistics Canada, the processor can also run aggregation queries against their own databases and transfer only summary counts, which would fall below PIPEDA's "personal information" threshold assuming the smallest cell sizes are large enough that they don't carry re-identification risk.

[118] Statistics Canada, "Data Ethics at Statistics Canada" (paper presented at the UNECE Conference of European Statisticians, November 2021), https://unece.org/sites/default/files/2021-11/S2_2_Canada_Data_ethics.pdf.
[119] OneTrust, "Quebec's Law 25;" Alation, "Quebec Law 25 Compliance Guide: Deadlines, Key Steps, Tools," Alation (blog), accessed April 12, 2026, https://www.alation.com/blog/quebec-law-25-compliance-guide/.

APPENDIX E

Occupational Composition by Province

Ontario has the most diversified canary cohort, with customer service representatives accounting for 38% and meaningful shares spread across software developers (15%), accounting clerks (14%), financial customer service representatives (13%), and software engineers (12%). A displacement signal in Ontario could arrive through multiple occupational channels simultaneously. The Atlantic provinces and Manitoba face a different situation. In PEI, 80% of the canary cohort consists of customer service representatives. New Brunswick is at 66%, Manitoba at 65%, and Nova Scotia at 56%. A single AI capability advance in customer service automation could affect the majority of the canary cohort in these provinces at once.



Source: Statistics Canada, Table 98-10-0593-01, 2021 Census of Population (25% sample). Workers aged 20-24 only. Total shown at right of each bar.

Figure 5. Occupation composition of the canary cohort by province (workers aged 20-24)

APPENDIX F

Key Decision-Makers and Counterparts

This appendix identifies the individuals whose decision-making authority or institutional role directly affects the implementation of this report's recommendations. Recommendation numbers in parentheses indicate where each individual's authority is most relevant.

Name and Title	Relevance to this Report
STATISTICS CANADA / TECHSTAT	
André Loranger *Chief Statistician of Canada*	Approves changes to survey instruments, including the LFS supplement frequency change (Rec 4) and CSBC module absorption (Rec 4). All data-sharing agreements with payroll providers require his sign-off.
Josée Bégin *Assistant Chief Statistician, Social, Health and Labour Statistics*	Oversees the Labour Force Survey. The digital platform supplement frequency request (Rec 4) goes through her directorate.
EMPLOYMENT AND SOCIAL DEVELOPMENT CANADA (ESDC)	
Patty Hajdu *Minister of Jobs and Families*	Co-chairs the FLMM for the 2025-2027 cycle. Authorizes Tier 3 escalation (convening provinces) under the signal-to-response protocol (Rec 5). Tables the means-testing working session at FLMM (Rec 6).
Paul Thompson *Deputy Minister, ESDC*	Senior bureaucratic authority over the Skills and Employment Branch, where the Labour Market Information Directorate sits.
Colette Kaminsky *Senior Assistant Deputy Minister, Skills and Employment Branch*	Direct authority over the Labour Market Information Directorate, which is TechStat's counterpart for the quarterly calibration sessions (Rec 5). Decides Tier 2 escalation (briefing the Minister).
INNOVATION, SCIENCE AND ECONOMIC DEVELOPMENT CANADA (ISED)	
Evan Solomon *Minister of Artificial Intelligence and Digital Innovation*	Statistics Canada reports into ISED. A signal from the Minister to the Chief Statistician carries significant institutional weight for prioritizing TechStat's work plan (Rec 1).
PAYROLL DATA PARTNERS	
Jim Lord *President, ADP Canada*	Initial outreach target for the voluntary payroll data partnership (Rec 4).

Dr. Nela Richardson *Chief Economist, ADP*	Oversees ADP's existing research partnerships, including the Stanford Digital Economy Lab collaboration that serves as the model for TechStat's proposed partnership.
David Ossip *CEO, Dayforce*	Initial outreach target. Dayforce has an existing Government of Canada relationship through the Phoenix pay system replacement.
INTERNATIONAL STANDARDIZATION	
Sam Manning *Senior Research Fellow, Centre for the Governance of AI*	Leading the multi-country effort to standardize AI usage reporting from frontier labs. Jonathan Barry's direct contact for communicating Canada's data requirements (Rec 3).
LABOUR MARKET INFORMATION COUNCIL (LMIC)	
Rebecca Webb *Executive Director, LMIC*	Contact for the joint technical briefing on the early warning system's data requirements proposed in Recommendation 3.
Suzanne Harbottle *Board Chair, LMIC*	Co-contact for the LMIC briefing. LMIC has standing relationships with labour market information officials in all 13 provinces and territories.